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Review

New Frontiers in AI-Nano Converged Platforms for Intelligent Diagnostics, Therapeutics, and Safety Evaluation

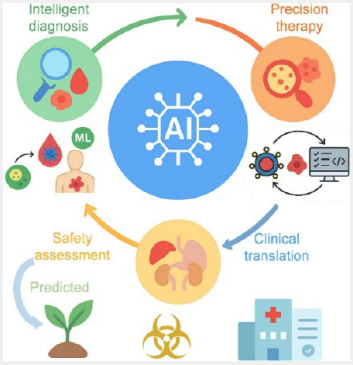
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ABSTRACT: The complexity of nanomedicine poses a significant challenge to traditional empirical methods. Machine learning (ML), with its ability to parse high-dimensional data and predict nonlinear interactions, is empowering AI nano fusion platforms, driving innovation in diagnostic and therapeutic evaluation paradigms. This review explores breakthrough applications of ML in core areas of nanomedicine, including intelligent diagnosis (such as ML enhanced nanosensors for high-precision noninvasive cancer early screening), precision therapy (such as closed-loop system driven rational design and delivery optimization of nanomedicine), safety assessment (such as interpretable AI prediction of biological and environmental toxicity of nanomaterials), and clinical translation (to address scientific challenges related to standardization, reproducibility, and regulatory). The paper provides an in-depth analysis of the key bottlenecks currently facing the transition from laboratory to clinical application, such as batch differences, dynamic interference, and regulatory lag. Based on recent research progress, a future path to achieve an intelligent closed-loop of "design diagnosis and treatment evaluation" is proposed, providing key insights for the development of the next generation of safe and efficient intelligent nanodiagnosis and treatment platforms.



KEYWORDS: *Machine learning, Nanomedicine, Intelligent diagnosis and treatment, Clinical translation*

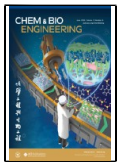
1. INTRODUCTION

Nanomedicine is committed to achieving precise diagnosis and treatment of diseases at the molecular and cellular scales, with its core advantage being the precise regulation of material physicochemical properties.[1−5] However, this advantage itself also introduces inherent complexity. The multiscale evolution process of nanoparticles in vivo, including initial protein crown formation, cell specific endocytosis, and ultimately drug release and metabolism at the subcellular level, together constitute a dynamic system full of nonlinear interactions.[6−9] The traditional "design synthesis test" paradigm heavily relies on experience and simplified models, making it difficult to comprehensively capture and predict key behaviors in this complex system.[10,11] This leads to a series of bottlenecks such as low efficiency in rational design, uncontrollable in vivo delivery, and difficulty in prospective evaluation of biosafety, ultimately delaying the transformation path of innovative nanomedicine solutions from the laboratory to clinical use.

The integration of artificial intelligence (AI), especially machine learning(ML), is providing a new paradigm for breaking through the complexity mentioned above. Its fundamental advantage lies in the ability to automatically learn and extract hidden, high-dimensional nonlinear mapping relationships from high-throughput experiments, multimodal imaging, and omics data.[12−17] This enables researchers to break through the limitations of traditional methods, manifested in three specific aspects. First, graph neural networks can be used to model the structure−activity relationship between surface chemistry of nanomaterials and dynamic protein adsorption.[18−22] Second, deep generative models can perform reverse engineering in a vast chemical space to construct nanostructures with specific functions.[23−28] Finally, reinforcement learning can help optimize individualized drug delivery strategies in simulated environments to adapt to different physiological microenvironments.[29−33] AI is no longer just a tool for auxiliary analysis but has become a core engine capable of understanding the complex language of nanobiological interfaces, predicting their behavior, and guiding rational design.

The integration of AI and nanotechnology shows great potential, but its translation into clinical and environmental applications still faces a series of common challenges, such as data scarcity and heterogeneity, which make it difficult to

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establish reliable structure−activity relationships; signal interference and matrix effects constrain the stability and accuracy of nano enhanced sensing in complex biological environments; the model's generalization ability is insufficient, limited by the inter batch differences and population diversity of the training data; and, the transformation gap from controlled experimental environments to real complex scenes. This review aims to systematically explain how AI Nano fusion leads the paradigm shift of intelligent diagnosis and treatment platforms (Figure 1). To this end, we will first analyze the fundamental bottlenecks of nanomedicine in terms of diagnostic sensitivity, therapeutic targeting, and system safety; Furthermore, it focuses on how AI can provide breakthrough solutions in the above-mentioned key areas through its powerful data parsing and pattern recognition capabilities, promoting the development of intelligent diagnosis, precision treatment, and safety assessment toward closed-loop automation; Ultimately critically examine the gaps that this technology must overcome in clinical translation, including data standardization, algorithm interpretability, and regulatory scientific adaptability, and outline a future roadmap for building an integrated intelligent system of "design diagnosis treatment evaluation", thereby laying a solid theoretical foundation for the substantial leap of next-generation nanomedicine from theoretical innovation to clinical benefits.

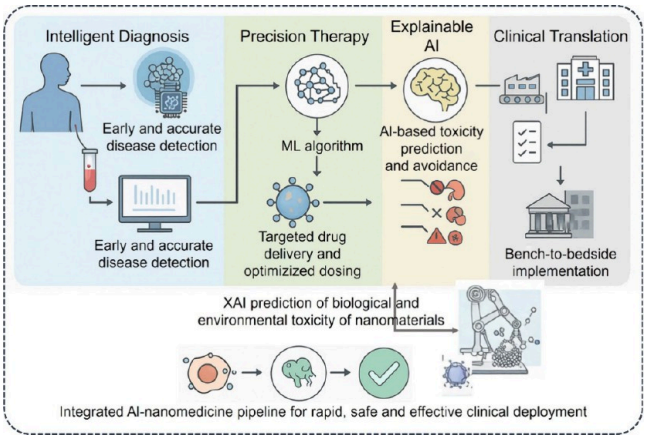


Figure 1. ML-driven Al nanofusion for next-generation nanomedicine.

2. ACCURATE DIAGNOSIS

2.1. Early Screening for Cancer

The urgent need for highly sensitive, specific, and convenient early cancer detection has driven a paradigm shift toward liquid biopsy, which is now being propelled by the integration of nanomaterial design and AI.[34−36] This paradigm of "nano enhanced sensing and ML decoding" faces fundamental limitations. A groundbreaking study has developed a gold nanomaterial (3D-EGN) with a three-dimensional hierarchical structure, which can directly address the two core challenges in urine sample detection, namely signal interference caused by Brownian motion and enrichment of weak target molecules.[37] The platform combined logical regression (LR) preliminary screening and convolutional neural network (CNN) multi

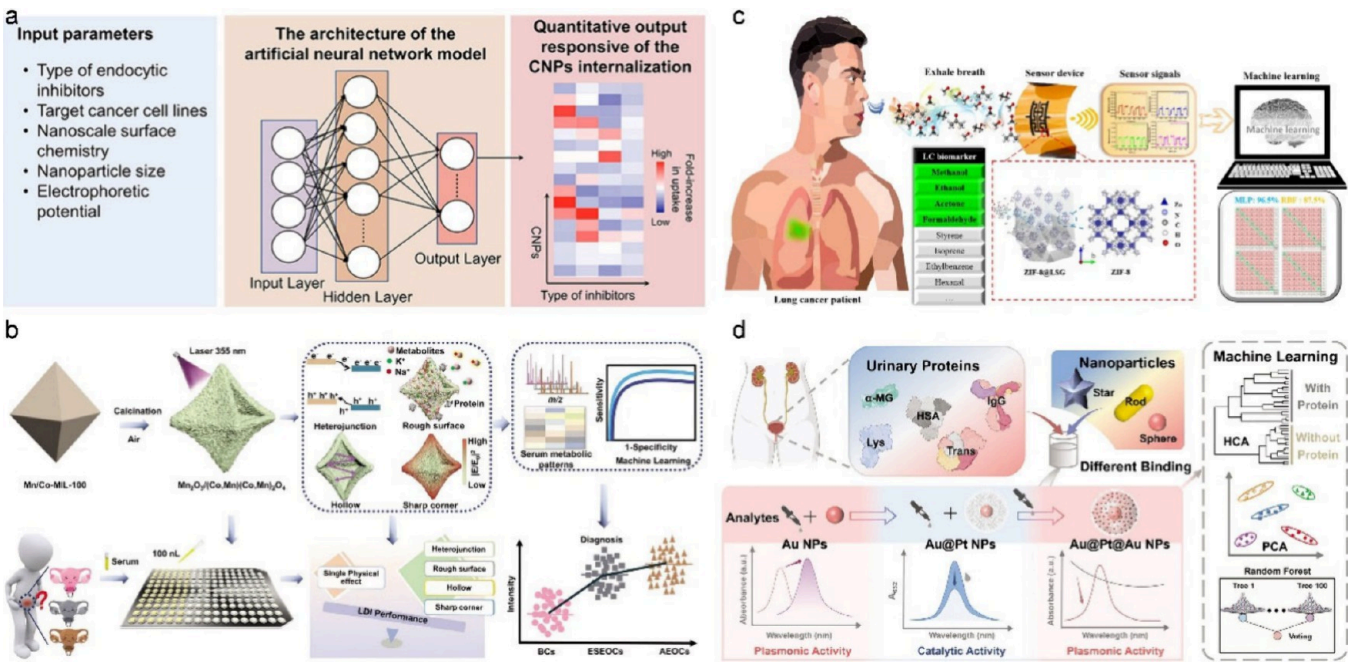


Figure 2. (a) Training of the artificial neural network model was done using the design parameters of eight CNPs to predict their cellular internalization. Reproduced with permission from ref 39. Copyright 2020, American Chemical Society. (b) Schematic workflow for the extraction of serum metabolic patterns by $Mn_2O_3/(Co,Mn)(Co,Mn)_2O_4$ (MO/CMO)-assisted LDI MS. MO/CMO with a heterojunction, rough surface, hollow interior, and sharp corner was prepared using Mn/Co-MIL-100 as a template. Reproduced with permission from ref 41. Copyright 2023, Wiley. (c) Concept of the ZIF-8@LSG chemoresistive sensor for VOC lung cancer biomarkers detection, enhanced with ML augmentation. Reproduced with permission from ref 42. Copyright 2024, The Royal Society of Chemistry. (d) Schematic illustration of the working principle of the "three-in-one" multifunctional nanoparticles-based sensor array for urine analysis. Reproduced with permission from ref 38. Copyright 2025, Wiley.

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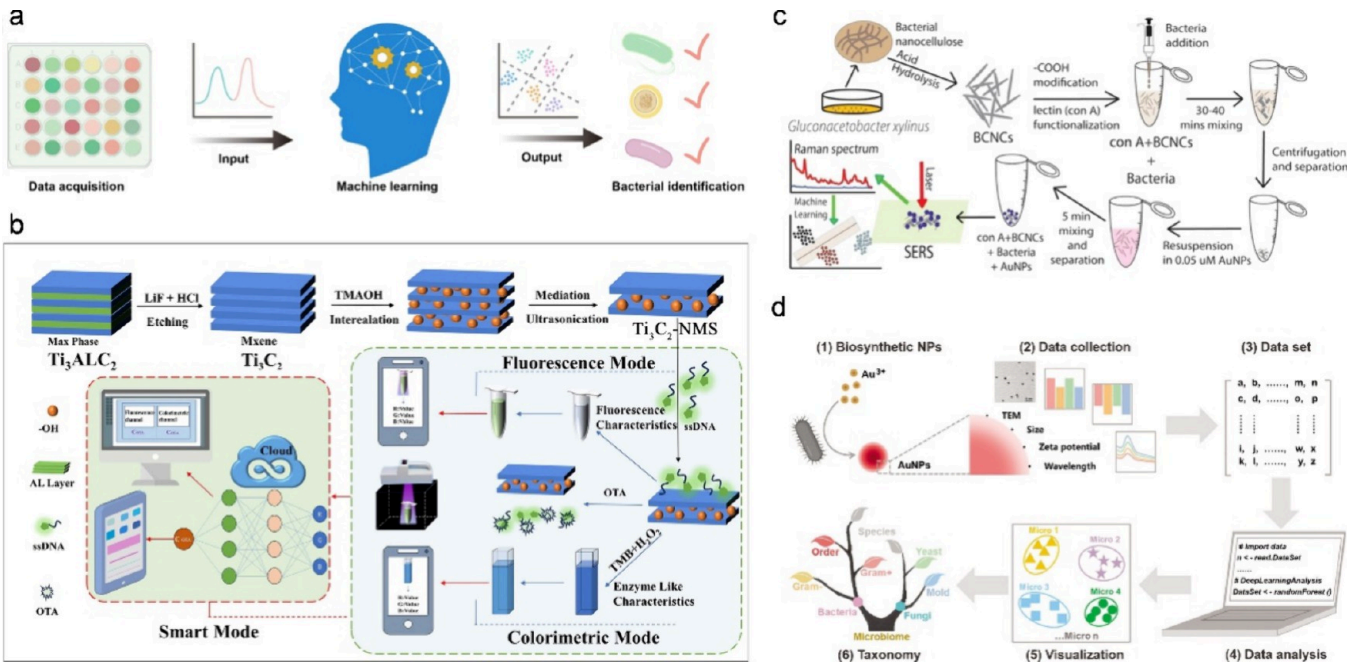


Figure 3. (a) Schematic illustration of bacterial identification based on the ML-aided bispecific metabolic monitoring platform. Reproduced with permission from ref 43. Copyright 2025, American Chemical Society. (b) The principle of detecting OTA by Ti3C2-NMS@ssDNA-FAM. Reproduced with permission from ref 45. Copyright 2025, Elsevier. (c) Schematic illustration of synthesis and functionalization of BCNCs, bacteria detection assay, SERS and machine-learning applications. Reproduced with permission from ref 46. Copyright 2022, American Chemical Society. (d) Schematic illustration of microbial taxonomic identification through biosynthetic AuNPs and ML. Reproduced with permission from ref 44. Copyright 2022, Wiley.

cancer classification, achieving an accuracy of 95.6% in five classifications of four types of cancer and normal samples, including pancreatic cancer and prostate cancer, and screened out specific metabolic markers. This strategy was further extended to the "three in one" urine protein analysis in the direction of body fluid detection (Figure 2d).[38] It constructs a sensor array by regulating the size/surface properties of nanoparticles, combined with ML feature extraction, and completes the qualitative and quantitative classification and comprehensive detection of bladder cancer related proteins within 10 min, significantly improving the efficiency of noninvasive diagnosis.

This paradigm has high universality in different sensing modes and biological samples. Early research achieved highly sensitive detection of biomarkers by constructing electrochemical sensing interfaces of green silver nanoparticles (GAgNPs).[39,40] This work not only achieved a detection accuracy of 98% for canine breast tumors using a random forest model but also further graded the malignancy of tumors using artificial neural networks (with an accuracy of 76%), systematically demonstrating that serum CA15−3 has better clinical predictive value compared to tissue MUC-1, providing new evidence for noninvasive liquid biopsy (Figure 2a).

Further advancing into omics-scale analysis, this strategy of "nano enhanced sensing ML decoding" has been deepened in the field of metabolomics. By designing MO/CMO heterojunction nanomaterials with unique concave octahedral structures and utilizing their physical properties of surface roughness and hollow sharp edges, the signal intensity of laser desorption ionization mass spectrometry can be significantly enhanced, enabling high-throughput detection of serum metabolites without complex preprocessing.[41] The researchers further employed sparse learning methods to successfully decode the malignant and benign metabolic fingerprints of ovarian cancer from enhanced mass spectrometry data, clearly revealing the synergistic mechanism between the physical effects of nanomaterials and ML feature extraction (Figure 2b). This approach has been extended to the field of gas-phase biomarker detection, and the development of a biomimetic ZIF-8@laser-induced graphene (LSG) hybrid sensor has achieved efficient capture of volatile organic compounds in exhaled breath of lung cancer patients.[42] By combining wavelet packet transform and principal component analysis for feature dimensionality reduction and using multilayer perceptron for classification, the system achieved a recognition accuracy of 96.5% for lung cancer, indicating that this fusion strategy is suitable for multiphase biological samples and laying a technical foundation for future Wearable Internet of Things diagnostic device (Figure 2c).

These studies demonstrate a new diagnostic logic that utilizes nanomaterials to generate rich and enhanced biochemical features and applies ML to mine accurate early disease classification from these high-dimensional data. This logic is currently being validated in various cancers and biological fluids.

2.2. Detection of Infectious Diseases

Rapid and accurate pathogen identification as well as antibiotic sensitivity testing are crucial in combating infectious diseases. The AI Nano fusion technology meets this demand by going beyond gene amplification and shifting toward functional phenotype analysis. Detection of infectious diseases is undergoing a paradigm transition driven by the collaborative innovation of nano biosensors and ML. A groundbreaking work has developed a dual color sustained luminescence nanoprobe that optimizes the efficiency of Foster resonance energy transfer through surface limited luminescence strategy,

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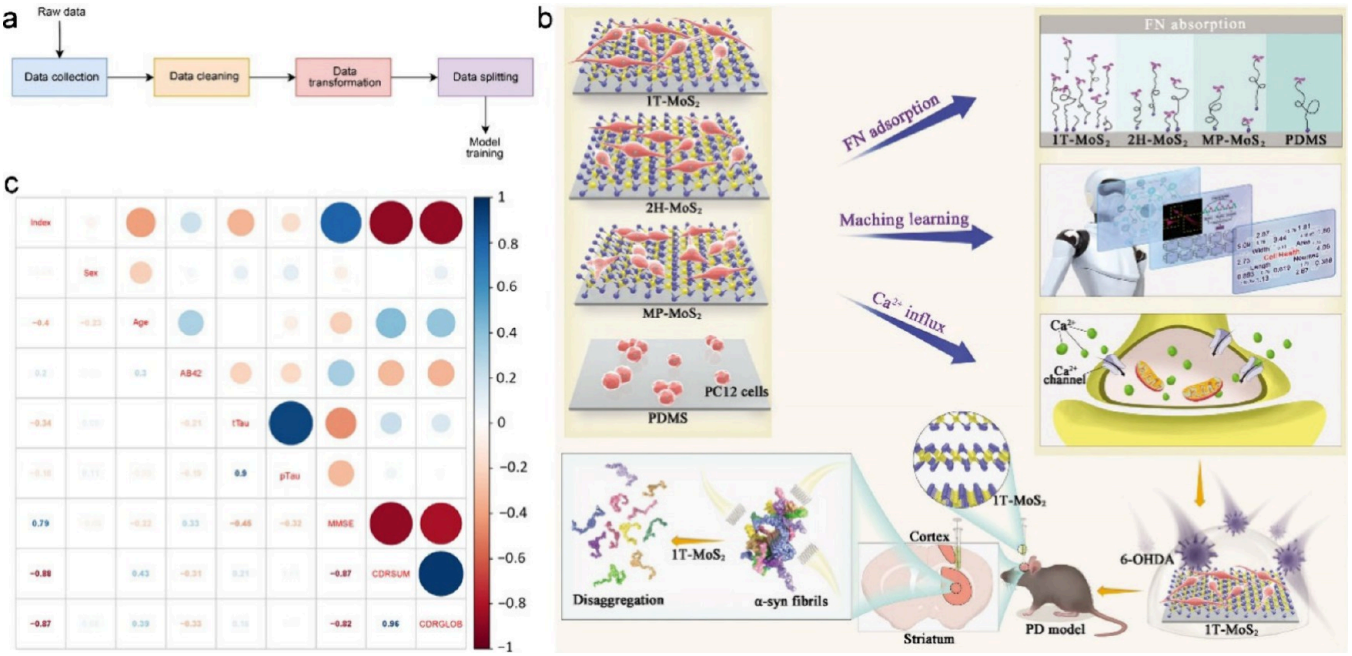


Figure 4. (a) Preprocessing of data flowchart. Reproduced with permission from ref 50. Copyright 2025, Springer Nature. (b). Preparation of 1T-MoS2, 2H-MoS2, and MP-MoS2 films and exploration of bionanoscale recognition underlying cell fate and therapy. Reproduced with permission from ref 51. Copyright 2021, Wiley. (c) Correlation analysis between prediction score and biomedical matrix. Reproduced with permission from ref 52. Copyright 2022, Elsevier.

specifically targeting bacterial metabolic enzymes and generating unique dynamic luminescent fingerprints (Figure 3a).[43] Coupled with ML analysis of the spatiotemporal enzyme activity patterns, this approach enables the 100% accurate identification of 5 pathogenic bacteria and the synchronous monitoring of antibiotic metabolic activity, thereby reducing the time required for drug sensitivity testing to 5 h. Based on similar logic, this metabolic monitoring paradigm has been further expanded into a universal microbial classification method that does not rely on gene amplification (Figure 3d).[44] It utilizes the size/SPR spectrum/zeta potential differences of gold nanoparticles (AuNPs) synthesized by microorganisms to construct a multilevel random forest classifier. Through the quantitative relationship between glucose concentration and SPR wavelength shift, it achieves 100% accuracy in the three-level classification of phylum, genera, and species, providing a new path for alternative gene amplification techniques.

The fusion mode fundamentally changes the way toxins are detected and improves the specificity of detection through multimodal sensing technology. At the same time, innovation in detection paradigms has also extended to small molecule targets, such as toxins. A study constructed a four mode optical sensor based on two-dimensional material MXene and improved the specific recognition of ochratoxin A through ligand modification (Figure 3b).[45] The innovative use of a dual channel fully connected artificial neural network (FCANN) to integrate multimodal signals and combined with mobile deployment to achieve rapid on-site detection has broken through the traditional laboratory dependence. This type of multimodal fusion strategy has been further deepened in bacterial recognition. Another work designed a functionalized SERS substrate for obtaining unlabeled spectral data of various bacteria.[46] The study used a support vector machine algorithm to analyze spectra and emphasized the key role of the standardized data preprocessing process in improving the model's classification robustness in complex samples, with an accuracy rate of 87.7%. This work, together with the aforementioned research on multimodal sensors, highlights the equal importance of data quality and algorithm processing in achieving accurate recognition, expanding the focus from simple sensor element design to the optimization of the complete "signal acquisition processing parsing" chain (Figure 3c).

This field is evolving from independent detection tools to integrated closed-loop intelligent systems. The direction of progress lies in the collaborative design of nanomaterials (for specific biological fingerprint recognition) and adaptive ML architectures (for robust multidimensional data fusion), with the aim to create a seamless platform for "identification, quantification, and efficacy evaluation" to manage infections and toxins.

2.3. Neurodegenerative Diseases

The early and accurate diagnosis of neurodegenerative diseases requires a comprehensive understanding of molecular dysfunction to comprehensive clinical manifestations. The AI Nano platform is opening up new avenues to achieve this goal by connecting nano sensing technologies at these different scales. By precisely designing nano sensing interfaces and deeply integrating ML algorithms, early diagnosis of neurodegenerative diseases is achieving multilevel breakthroughs from single molecule detection to pathological mechanism analysis, and then to clinical integrated diagnosis.[47−49] This progress is first reflected in the highly specific quantification of key neurotransmitters. A study developed an electrochemical sensor based on molybdenum disulfide quantum dots/carbon nanocomposites, which significantly improved the redox current response of dopamine by enhancing the catalytic activity of molybdenum sulfur edge sites (Figure 4a).[50] In the face of signal interference from structurally similar substances

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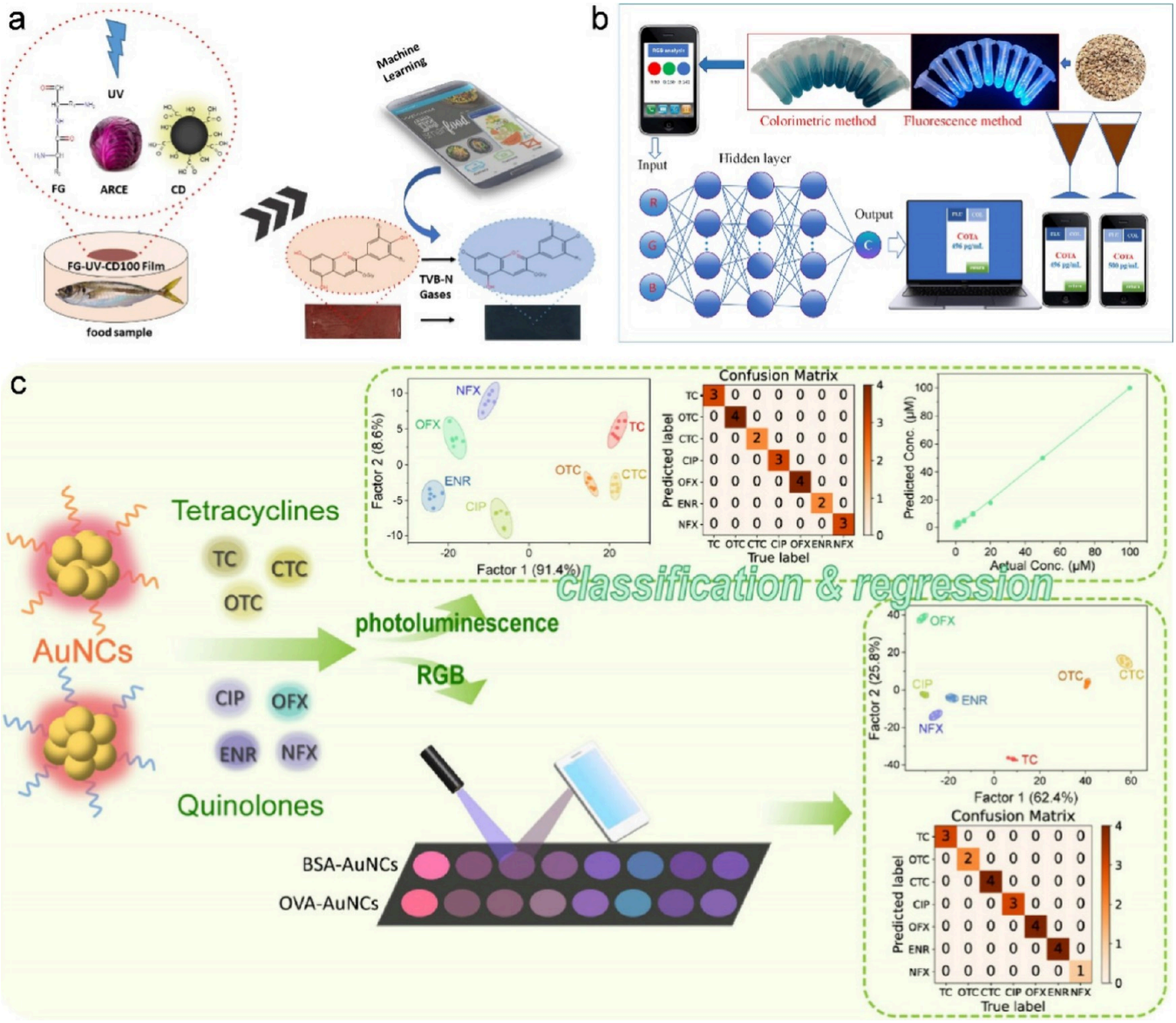


Figure 5. (a) Schematic representation of the proposed system. Reproduced with permission from ref 60. Copyright 2023, Elsevier. (b) The structure of the FCANN model and its application diagram in practical detection of OTA. Reproduced with permission from ref 45. Copyright 2025, Elsevier. (c) Schematic illustration of the photoluminescent sensor array based on AuNCs for antibiotics identification assisted by ML and test paper. Reproduced with permission from ref 63. Copyright 2024, Elsevier.

(such as ascorbic acid and uric acid) in actual biological samples, this work innovatively introduces artificial neural networks to decouple and analyze overlapping signals, ultimately achieving high-precision dopamine detection in human serum with a detection limit of 0.6 μ mol · L $^{-1}$ (R^2 > 0.98), effectively overcoming the inherent bottleneck of insufficient selectivity in traditional electrochemical methods.

In addition to quantitative analysis, this method lays the foundation for revealing the mechanisms of nanobiological interactions related to pathology. The above precise detection paradigm at the molecular level lays the foundation for further revealing the mechanism of interaction between nanomaterials and pathologically related proteins. Subsequent research will focus on the correlation between molecular assembly and pathological processes, systematically exploring the differential regulation of fibronectin adsorption behavior by the topological structure of different crystal phases (1T phase and 2H phase) of single-layer MoS_2 (Figure 4b).[51] By combining random forest and structural equation modeling for multivariate analysis, this study revealed that this specific protein adsorption behavior further affects the depolymerization process of alpha synuclein and the survival status of nerve cells, providing a new theoretical basis and design principles for designing neuroprotective nanotherapies based on material surface−interface engineering.

Ultimately, the technology matured through direct integration with complex clinical information capable of converting spectral patterns into diagnostic scores. Based on the profound insights accumulated from molecular recognition to mechanism analysis, diagnostic strategies are further evolving toward integrating clinical information and serving practical diagnosis and treatment. A clinical translational study constructed a surface enhanced Raman spectroscopy CNN diagnostic platform (Figure 4c).[52] The platform is no longer limited to

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a single biomarker but utilizes surface enhanced Raman spectroscopy to directly capture diverse pathological features such as amyloid beta aggregates in cerebrospinal fluid and deeply decodes complex spectral fingerprints through CNNs. The model successfully established a strong correlation between spectral features and clinical dementia scores, achieving an overall diagnostic accuracy of up to 92% for early stages of Alzheimer's disease in a double-blind trial, marking an important step in AI nano fusion diagnosis from principle validation to addressing clinical complexity and heterogeneity challenges.

This process demonstrates a powerful multiscale strategy from precise sensing at the molecular level, to mechanism based material design, and to comprehensive diagnosis at the clinical level. This lays a coherent framework for constructing a closed-loop system that connects molecular detection with disease modeling and diagnosis.

2.4. Portable POCT System

Point-of-Care Testing (POCT) refers to analytical devices and systems designed to perform diagnostic tests at or near the site of patient care, providing rapid results to guide clinical decisions. POCT system is undergoing a paradigm upgrade driven by nano functional materials and embedded ML.[33,53−56] Its core goal is to solve the long-standing triangular contradiction in field applications, namely the tense relationship between complex environmental interference, multimodal signal analysis requirements, and limited terminal computing power.[37,57−59] From single mode analysis to multidimensional information fusion, from in vitro detection to continuous physiological monitoring, this series of research clearly shows the development path of technology upgrading.

Early innovation provided a solution to address the fundamental challenge of environmental variables on the detection stability. By constructing a fish gelatin carbon dot composite colorimetric film and training it on an anti-interference data set covering multiple light sources, devices, and perspectives, a study successfully embedded a random forest model into a smartphone application, achieving rapid offline analysis of amine decay markers with an on-site accuracy of up to 99.6%, establishing a new paradigm for lightweight food safety detection (Figure 5a).[60] This work confirms that compensating for environmental noise through algorithms can effectively compress laboratory analysis processes for mobile devices.

Following the direction of mobile integration and signal fusion, subsequent research has further improved the specificity and reliability of complex sample detection. By combining a four modal sensor based on Ti_3C_2 MXene with a dual channel fully connected neural network and utilizing ssDNA functionalization to enhance specific recognition, high-performance detection of ochratoxin A was achieved.[45] This strategy fully taps into the collaborative potential of multi-modal sensing signals, successfully transferring laboratory level analysis to smartphone platforms, marking a key shift in POCT from relying on a single signal to comprehensively utilizing high-dimensional biochemical information for intelligent decision-making (Figure 5b).

At the same time, the scope of POCT is expanding from in vitro sample analysis to real-time interpretation of continuous physiological signals, which puts higher demands on the wearable performance of sensing interfaces and the real-time processing capability of algorithms. To address this challenge, relevant research has undergone two stages of systematic evolution. In the initial research, printed flexible nano-electrodes were used to collect electromyography and CNNs were used to decode muscle movement signals. Subsequently, research further innovated by constructing a fully printed wireless system using solderable graphene circuits, optimizing multichannel electromyographic feature extraction using deep learning, and ultimately achieving millisecond level recognition of multiple finger movements.[61,62] This development path effectively breaks through the bottleneck of traditional electrodes in long-term comfort and signal fidelity, laying a solid foundation for the practical application of wearable diagnostic technology. The analysis logic based on "smart material embedded algorithms" has been proven to be equally applicable to the field of environmental monitoring. For example, a photoluminescence test strip array based on gold nanoparticle clusters, combined with ExtraTrees regression model, has achieved high-precision quantification of multiple antibiotics; By using smartphones for RGB analysis, this work transforms the traditional laboratory quality control process into a portable and fast on-site inspection tool (Figure 5c).[63]

Looking at the current progress, the common breakthroughs are mainly reflected in three interrelated aspects, namely, environmental robustness design for complex field applications, lightweight algorithm optimization for terminal deployment, and nanobiological interface innovation aimed at improving detection sensitivity and selectivity. However, moving toward large-scale clinical and field applications still faces significant challenges, including batch differences in multimodal sensors, signal drift in complex physiological scenarios, and interference from actual sample matrices on AI models. The future development relies on developing adaptive calibration algorithms (such as online incremental learning to combat device aging), exploring neural morphology computing hardware to reduce power consumption, and establishing standardized nanomaterial sensing libraries to ensure detection consistency. Through deep interdisciplinary integration, it will ultimately promote the formation of a highly autonomous and reliable "sample in result out" intelligent diagnostic closed-loop system.

To address the fundamental tradeoff between nano enhanced sensitivity and matrix induced signal variability, future diagnostic platforms must integrate adaptive ML frameworks. Domain adaptation and transfer learning can recalibrate models trained under controlled conditions to operate reliably in diverse physiological and environmental contexts. Meta learning algorithms with the ability to learn how to learn can quickly adapt to new distractor features or patient-specific baselines with minimal additional data. In addition, incorporating a physiology based pharmacokinetic (PBPK) model as prior knowledge in Bayesian neural networks can enable the model to predict and correct for foreseeable in vivo signal disturbances. The synergistic effect of powerful nano-material design and these adaptive computing strategies is crucial for translating laboratory level accuracy into real-world diagnostic reliability.

3. INTELLIGENT THERAPY

3.1. Anticancer Nanodrugs

The development paradigm of anticancer nanomedicine is undergoing a fundamental shift, shifting from traditional empirical screening to an intelligent cycle of "computational

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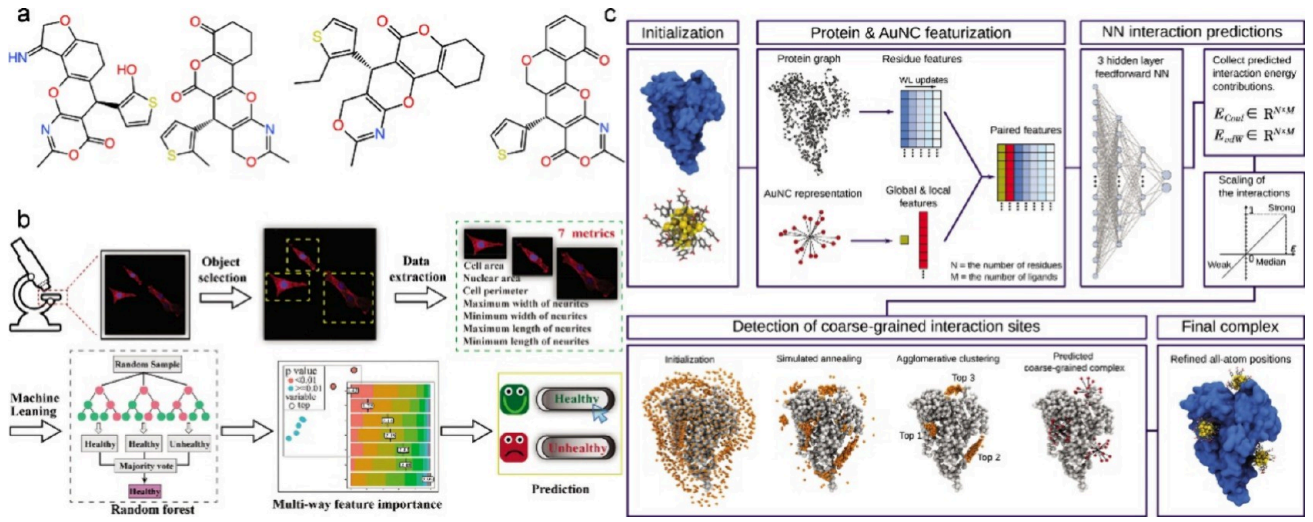


Figure 6. (a) Four structures were selected from 6600 optimized episodes, their binding energies were 9.2, 9.4, 9.4, and 9.4 kcal/mol, the SA scores were 0.6018, 0.6282, 0.6491, and 0.6298, and the QED scores were 0.7994, 0.7591, 0.7745, and 0.7710. Reproduced with permission from ref 66. Copyright 2025, Elsevier. (b) Flowchart showing the cell image processing analysis routine. Reproduced with permission from ref 51. Copyright 2021, Wiley. (c) The GraphBNC framework consists of four steps: featurization, interaction energy predictions, coarse-grained annealing placement, and refinement to all-atom positions. Reproduced with permission from ref 64. Copyright 2024, Wiley.

design experimental verification".[64] This transformation aims to systematically address core challenges such as tumor heterogeneity, delivery efficiency, and off target toxicity through data-driven approaches, and its development trajectory is reflected in the full chain of innovation in design, prediction, and clinical translation.

Early research laid the foundation for computer-aided design. By constructing a dual targeted (folic acid/small molecule co modified) nanoparticle library and applying sparse regression models, the structure−activity relationship between surface chemical properties and cancer cell uptake was quantitatively analyzed for the first time.[65] This work not only reveals the synergistic mechanism between ligands but also achieves the prediction of personalized targeting effects, providing a new quantitative rule-based approach to overcome tumor heterogeneity. On this basis, the concept of computational design has further evolved into an active generation strategy. Starting from a specific lead compound, a large number of anti lung cancer molecular fragments were iteratively generated using reinforcement learning algorithms, and systematically integrated with pH responsive inorganic polymer nanocarriers.[66] This study has achieved a "design manufacture" integrated closed-loop from molecular generation to the construction of nano delivery systems, marking a shift in the role of AI in nanomedicine development from auxiliary analysis to active creation (Figure 6a).

Accurately predicting the interaction between nanomaterials and biological systems is a key prerequisite for a successful intelligent delivery. For this purpose, a graph neural network GraphBNC has been developed, which can efficiently predict the interaction strength and binding sites between gold nanoclusters and blood proteins based solely on single component molecular dynamics data.[64] By combining the Monte Carlo structural search for atomic scale optimization of the composite, this framework significantly reduces the systemic toxicity risk of large-sized nanoparticles during in vivo delivery (Figure 6c). Similarly, insights into the inherent physicochemical properties of nanomaterials provide guidance for rational design. By combining random forest and structural equation modeling, the study revealed that the crystal phase topology of single-layer molybdenum disulfide affects the intrinsic pathway of cancer cell viability by regulating fibronectin adsorption, providing a clear physical and chemical blueprint for surface engineering based biological effect regulation of nanomaterials (Figure 6b).[51] Ultimately, the value of intelligent design needs to be validated in complex disease models. The key synthesis parameters and drug release kinetics of folate targeted hybrid nanocarriers were successfully optimized by integrating an adaptive neural fuzzy system and a hybrid model of extreme gradient boosting tree.[67] This strategy achieved high encapsulation efficiency and significant inflammation inhibition in a rheumatoid arthritis model, demonstrating the strong potential of computational optimization frameworks in the development of cross disease nanomedicines.

Despite the promising prospects of these computational frameworks, there still exists a significant "reality gap" between the simplified simulations used for training and the complexity actually transmitted in vivo. Reinforcement learning models used for molecular generation typically operate in confined chemical spaces and lack integration with dynamic physiological barriers. As shown in Table 1 comparison, although predictive models like GraphBNC are very powerful in protein interaction screening, they may overlook heterogeneous tumor blood vessels, variable interstitial pressure, and dynamic immune cell interactions, which play a crucial role in the fate of nanocarriers. This gap highlights the necessity of computational models that can clearly consider multiple aspects of the tumor microenvironment and have fewer constraints in the parameter space.

3.2. Antibacterial Materials and Antibiotics

The paradigm of antibacterial nanotechnology is undergoing a profoundly intelligent upgrade. Its core feature is to transform from the optimization of isolated links to the construction of an integrated closed-loop system of "material prediction-biological perception-accurate detection-effect verification", so

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Table 1. Comparison Table of ML Paradigms Mentioned Above

Algorithm	Resolve pain points	Clinical Interface	In Vivo Complexity Gap	Priority for Next-Generation Modeling	Ref
R L	Efficient generation of anti lung cancer molecules	Nanocarrier encapsulation verification	Ignores synthetic feasibility, pharmacokinetics, and dynamic tumor pathophysiology	Integrate RL with physiologically based pharmacodynamic (PBPK-PD) models and high-throughput organoid screening	66
Graph neural network and Monte Carlo	Prediction of nano protein interaction	In vivo toxicity pre screening	Lacks dynamic protein corona evolution, competitive binding in multiprotein plasma, and shear stress effects in circulation	Develop dynamic GNNs trained on time-resolved proteomics data and integrate fluid dynamics simulations	64
ANFIS-XGBoost	Synthesis parameters—Release kinetics mapping	Optimization of therapeutic effect of inflammation model	Does not capture in vivo bioavailability, immune clearance, and off-target organ accumulation	Employ multiobjective Bayesian optimization balancing in vitro potency with predicted in vivo safety and PK profiles	67

as to systematically reshape the diagnosis and treatment mode of infectious diseases.

The starting point of this intelligent upgrade lies in using ML to subvert the traditional experiential design mode of antibacterial materials. Early research demonstrated the potential of computational guided rational design by establishing predictive models for the antibacterial activity of block copolymers.[68] This work utilizes a decision tree algorithm combined with SHAP interpretability analysis to quantitatively reveal the contribution weights of key molecular descriptors such as "c log P", and proposes new polymer design principles based on the principle of hydrophilic oleophilic balance, providing an interpretable computational blueprint for synthesis. Subsequently, the intelligence of material design was further advanced to the nanoscale. By applying an extreme random tree model, the study successfully correlated nanoscale structural parameters such as doping concentration and particle size with bacterial survival rate based on absorbance quantification, thereby eliminating the subjectivity of traditional inhibition zone determination methods and achieving standardized and predictable design of antibacterial effects (Figure 7a).[69]

The development of precise and rapid detection technology that matches high-performance antibacterial materials based on rational design has become a key step in verifying their efficacy and achieving closed-loop applications. In this direction, researchers have developed a fluorescence sensor array based on dual channel carbon quantum dots.[70] The sensor, combined with a support vector machine, decodes the response fingerprint of tetracycline antibiotics and successfully achieves accurate classification in complex matrices such as river water and milk, providing an efficient solution for monitoring antibiotic residues in the environment and food. Similar intelligent sensing logic has also been applied to the detection of infection related biomarkers. By constructing a CuO ZnO p-n heterojunction sensor array and combining support vector machine and multiple linear regression algorithms to analyze the response dynamics of volatile organic compounds, the problem of multicomponent gas cross response has been effectively solved, providing a new tool for respiratory based infection diagnosis.[71]

Despite significant progress, building a complete intelligent antibacterial diagnosis and treatment chain still faces multiple challenges, mainly including the scarcity of high-quality training data, interference from complex real-world environments in sensing signals, and the conversion gap between laboratory predictions and clinical efficacy verification. Looking ahead, as shown in Table 2, research in this field should prioritize the development of transfer learning frameworks suitable for small sample scenarios to alleviate data bottlenecks.. At the same time, it is necessary to combine microfluidic organ chip technology to simulate the human infection microenvironment, providing an in vitro model that is closer to physiological conditions for the validation of materials and algorithms. Ultimately, through the integration of reinforcement learning and other algorithms, we are expected to explore an autonomous optimized diagnosis and treatment path from intelligent material synthesis to adaptive drug delivery and to real-time efficacy monitoring, achieving global closed-loop intelligent regulation of antibacterial nanotechnology.

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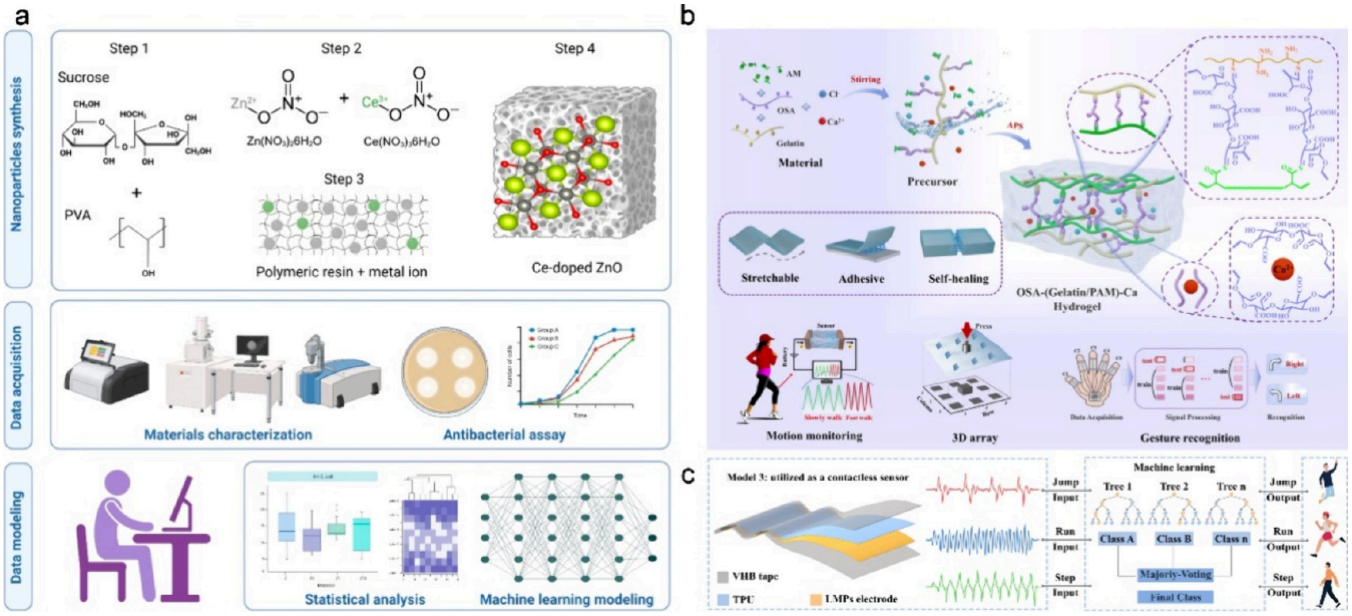


Figure 7. (a) Diagram of the process performed for the synthesis, characterization, application, and modeling with ML. Reproduced with permission from ref 69. Copyright 2023, American Chemical Society. (b) Schematic synthesis, functions, and applications of hydrogel. Reproduced with permission from ref 75. Copyright 2024, Elsevier. (c) Schematic structure of the contactless sensor, ML algorithm of random forest for motion patterns recognition, Diagram of different movement patterns. Reproduced with permission from ref 76. Copyright 2023, Elsevier.

Table 2. Comparison Table of Scientific Breakthroughs in Developing Transfer Learning Frameworks Suitable for Small Sample Scenarios

Key technological contributions	Clinical Interface	Ref
SHAP analysis of the antibacterial mechanism of "c log P"	Principles of rational design for polymers	68
Standardized prediction of absorbance replacing ZOI	Evaluation of antibacterial efficacy of nanoparticles	69
Dual channel fluorescence array and SVM complex matrix recognition	Environmental/food antibiotic pollution monitoring	70
SVM-MLR collaborative cracking of sensor cross response	Monitoring of volatile compounds in wound infection	71

3.3. Tissue Engineering and Regenerative Medicine

The fields of tissue engineering and regenerative medicine are integrating intelligent flexible sensing and ML algorithms to achieve real-time and accurate feedback and intervention in tissue repair processes, thereby promoting the paradigm evolution of rehabilitation models from passive monitoring to active interaction and adaptive regulation.[41,72−74]

Building a biological interface that can work stably for a long time is the primary foundation for achieving this goal. Relevant studies have successfully prepared double network conductive hydrogels with dynamic bonding and self-healing characteristics by covalently connecting sodium oxide alginate to gelatin polyacrylamide network (Figure 7b).[75] The three-dimensional array sensor integrated in this material can simultaneously capture strain and pressure signals to decode joint motion trajectories. By utilizing the fusion strategy of decision tree, support vector machine, and K-nearest-neighbor algorithm, the system has achieved high-precision classification of gesture actions. Of particular importance is that the self-healing material exhibits minimal degradation in sensory performance after damage repair, providing an ideal interface with high environmental robustness for rehabilitation monitoring that requires long-term and continuous use. On the basis of achieving stable perception, the intelligence of rehabilitation systems is further developing toward the integration of multimodal physiological signals and real-time human−computer interaction. Unlike single signal monitoring, another study focuses on developing a multifunctional sensing platform that can synchronously collect joint mechanical strain, electrocardiogram signals, and surface electromyography signals.[76] The liquid metal thermoplastic polyurethane ultra stretched film prepared based on electrospinning technology maintains stable electrical properties under extremely high strain (Figure 7c). By using decision tree algorithm to process multimodal signals in real time, the system can accurately identify various motion modes such as walking, running, jumping, etc. with extremely low latency, and has been successfully applied to immersive electromyography interactive game control, marking the progress of rehabilitation training toward personalized and highly participatory active human−machine collaboration mode.

Although the two technological paths mentioned above have different focuses, they jointly confirm the core value of deep

Table 3. In Depth Analysis of Common Challenges in Clinical Translation

Challenge Dimension	Environmental stability	Signal coupling	Individualized adaptation
Wang's research performance[75]	Conductivity drift caused by immersion in body fluids	Strain/pressure cross interference	The gesture library does not cover pathological movement patterns
Ye's research performance[76]	Sweat erosion resistance fluctuation	EMG - asynchronous motor timing	Dynamic changes in exercise characteristics during rehabilitation period

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collaboration between advanced materials, multimodal sensing, and intelligent algorithms. The former ensures the reliability of long-term monitoring through material innovation, while the latter opens up the possibility of dynamic interaction through information fusion and real-time computing. However, as summarized in Table 3 for the comparison of common challenges in clinical translation, common bottlenecks such as physiological and environmental interference, motion artifact coupling, and lack of personalized adaptive capabilities still need to be overcome in order to achieve widespread clinical application translation. It is necessary to develop more advanced model frameworks for future research such as using graph convolutional networks to construct mapping relationship models from "material microstructure changes" to "macroscopic signal attenuation". By combining transfer learning techniques, such models can quickly adapt to individualized rehabilitation characteristics of different patients, ultimately promoting the formation of a closed-loop intelligent rehabilitation system based on a "real-time perception intelligent evaluation adaptive adjustment".

The key to the combination of AI and nanotechnology lies in the matching between algorithm selection and material data features. As shown in Table "4", different ML methods have their own expertise in nanomaterial data types or clinical applications due to their inherent characteristics. CNNs are adept at processing high-dimensional data with spatial correlations, such as surface enhanced Raman spectroscopy, nanoparticle microscopy images, or metabolomics heatmaps, and are therefore widely used in intelligent diagnosis based on nano enhanced sensing. In contrast, Random Forest (RF) and Gradient Boosting algorithms have strong robustness against overfitting, can effectively handle mixed types of tabular data (such as size, zeta potential, functional group descriptors, etc.), and are often used to construct quantitative structure−activity relationship models. Graph neural networks (GNNs) naturally fit the topological structure of nanomaterials and their interaction networks (such as protein crown formation) and perform outstandingly in predicting the behavior of nano-biological interfaces. However, the lack of interpretability in many deep learning models poses obstacles to their clinical translation, highlighting the importance of introducing interpretability techniques such as SHAP in the safety assessment. It can be seen that there is no universal "optimal" ML method; The selection of algorithms must be closely integrated with the specific data types generated by nanotechnology research, while taking into account the practical needs of the transformation scenario (such as interpretability, real-time performance, etc.).

4. SAFETY ASSESSMENT

4.1. Nanobiological Interface Mechanisms

Analyzing the dynamic complexity of protein crown formation and nanobiological interface interactions, in order to go beyond traditional toxicological end point observations and achieve mechanism prediction and active design for biosafety.[77−79]

The starting point of this field is to establish an interpretable relationship among the physicochemical properties of nanomaterials, dynamic protein crowns, and their in vivo fate. Early research proposed a P/NP ratio strategy to regulate competitive binding of plasma proteins, combined with a fully automated platform for deep plasma proteomic

Table 4. Comparison of Key ML Algorithms in the AI-Nano Fusion Platform

Algorithm	Best Applicable Data Type	Core strengths	Main limitations	Typical Applications of AI Nano	Consideration of clinical translational maturity
Convolutional Neural Network	Image, spectrum, spatial data	Automatic feature extraction, high-dimensional data processing	Requires a large amount of data \ Poor interpretability	Medical imaging analysis, SERS/Raman spectroscopy decoding	High (widely validated), but model generalization needs to be addressed
Graph Neural Network	Graph structured data (molecules, interaction networks)	Directly modeling relationships and topological structures	Computational complexity and sensitivity to graph quality	Nanomaterial design, protein crown prediction, drug target prediction	Medium (rapidly developing), requiring more experimental verification
Random Forest/ Gradient Boosting	Structured/tabular data	Anti overfitting, providing feature importance, fast training	The performance of high-dimensional sparse data decreases	QSAR modeling, toxicity prediction, sensor signal classification	High (robust, easy to explain, commonly used for early screening)
Reinforcement Learning	Sequential decision-making problem	Optimize dynamic and multi-step decision-making processes	High exploration cost and unstable training	Optimization of nanomedicine delivery scheme and control of synthesis process	Low to moderate (during the concept validation phase, there is a significant gap between the simulated environment and the actual situation)
Explainable AI	Any black box model output	Enhance model transparency and trustworthiness	Possible increase in computational cost, explanation itself needs to be verified	Possible increase in computational cost, explanation itself needs to be verified	High demand

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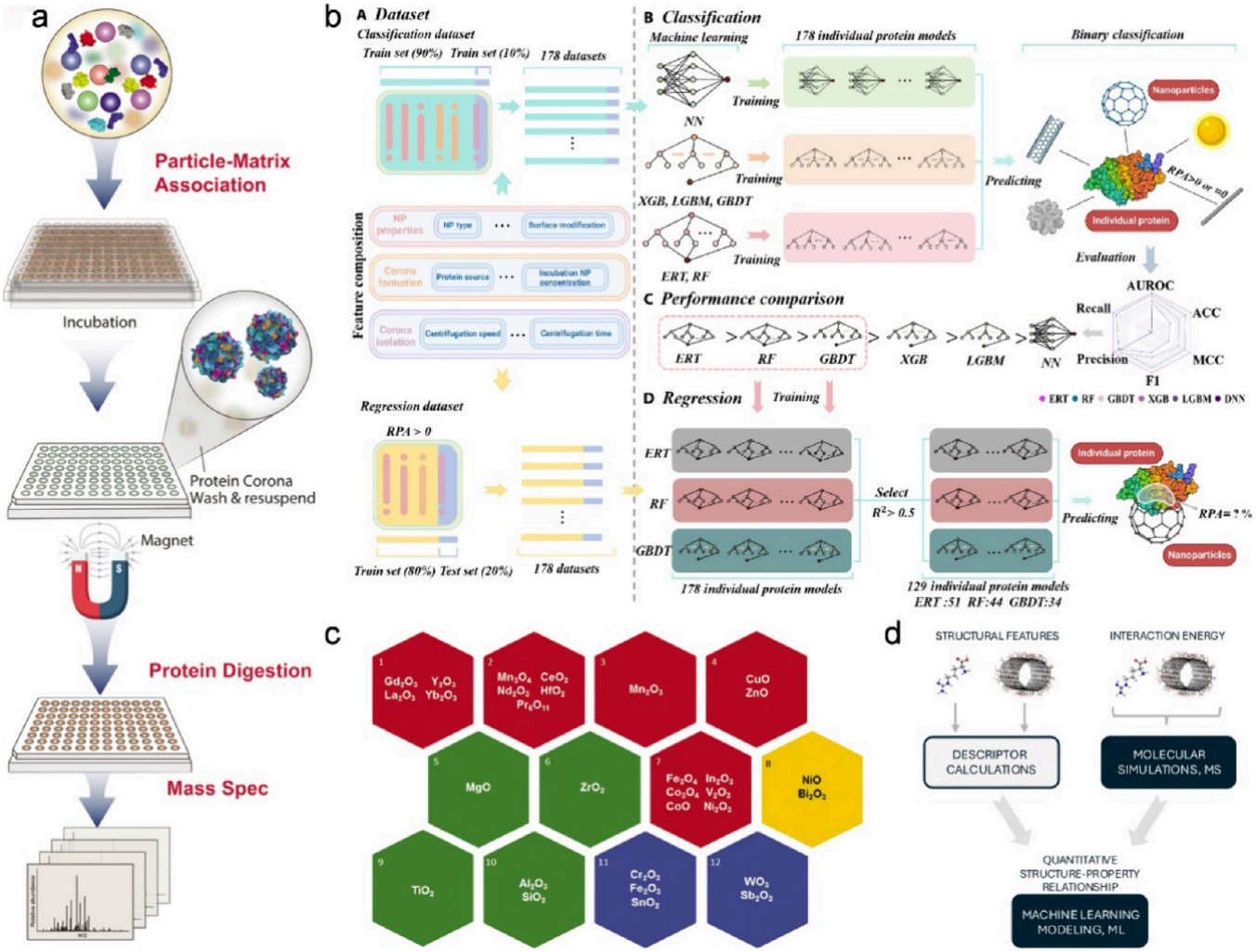


Figure 8. (a) Proteograph workflow (LC-MS/MS and downstream data processing) schematics. Reproduced with permission from ref 80. Copyright 2022, Wiley. (b) The overall experimental flow framework diagram. Reproduced with permission from ref 81. Copyright 2024, American Association for the Advancement of Science. (Research, 2024). (c) Schematic of similarity in toxicity between MeOx nanoparticles by applying the SOM clustering technique. Reproduced with permission from ref 83. Copyright 2018, The Royal Society of Chemistry. (d) Scheme of the hybrid, meta-modeling approach for predicting nano−bio interaction. Reproduced with permission from ref 84. Copyright 2025, The Royal Society of Chemistry.

identification, and for the first time correlated the dynamic composition of protein coronations with the in vivo biological distribution of nanoparticles through unsupervised learning.[80] This work provides a key mechanistic basis for understanding the in vivo behavior of nanocarriers and identifies potential pathways for regulating their biological distribution through interface engineering design (Figure 8a).

Based on the understanding of the above mechanisms, the research focus quickly shifted to developing predictive computing tools. By constructing a ML framework that integrates the dual tasks of classification and regression, relevant research has achieved synchronous prediction of protein adsorption probability and its relative abundance.[81] This model, combined with SHAP interpretability analysis, quantitatively reveals the contribution of key features such as "unmodified surface", and is deployed as an online platform to achieve rapid virtual screening of protein crown behavior (Figure 8b). However, the generalization ability of such models is limited by the scarcity of high-quality standardized data. Previous studies have attempted to use multivariate ML to analyze the nonlinear relationship between protein crown composition and cell binding, but its extrapolation ability is limited due to issues such as feature collinearity.[82] This data bottleneck was clearly identified in subsequent work; for example, in a study involving 178 proteins, there were only 652 shared data points, highlighting the urgency of building a broader data set.

To overcome the dual limitations of data and generalization, subsequent research follows two complementary paths. One path is dedicated to integrating cross species and multi type experimental data to explore universal patterns. By combining self-organizing mapping and supervised learning to analyze cross species toxicity data of 184 nanoparticles, the study successfully identified common toxicity patterns of metal and oxide nanoparticles and revealed sensitivity differences between different model organisms, effectively filling knowledge gaps with existing data (Figure 8c).[83] The other path turns to computational simulation to generate high-dimensional theoretical data to supplement the experimental deficiencies. A hybrid framework combining molecular dynamics and ML has been developed to predict the adsorption free energy of nano surfaces with biomolecules

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based solely on their topological and chemical descriptors, reducing computational costs by 90% while maintaining reasonable accuracy (validation set R^2 = 0.70) (Figure 8d).[84] Although this model still faces challenges in simulating complex solution environments, such as the effects of zeta potential drift, it represents an important attempt to predict the behavior of nanobiological interfaces from a first principles perspective.

In summary, the comparison table of common challenges and cracking strategies in this section shows that these studies collectively outline three main breakthrough directions in this field, namely, developing models that can capture dynamic processes, improving model interpretability, and constructing modeling frameworks that integrate multiscale information (Table 5). Looking ahead to the future, developing more advanced graph neural networks to handle complex correlations, integrating in situ characterization data to calibrate models, and ultimately constructing causal inference chains from atomic scale adsorption energy to macroscopic cellular reactions will be key to achieving active design and accurate prediction of nanomaterial safety.

Table 5. Comparison Table of Common Challenges and Cracking Strategies in this Chapter's Work

Challenge	Representative response plan	Limitations and breakthrough directions	Ref
Data heterogeneity	Dual task transfer learning	Graph neural network cross protein generalization	81
Interpretability of mechanism	SHAP analysis of unmodified surface contribution	Visualization of dynamic adsorption energy trajectory	81
Cross scale correlation missing	Molecular dynamics ML hybrid framework	In situ cryoelectron microscopy data fusion	84

4.2. From a Single End point to a Complex System

The field of environmental and ecological risk assessment is undergoing a profound methodological transformation. The research on toxicity prediction based on ML has clearly evolved from evaluating single biological end points to simulating population level effects and from relying on static toxicity indicators to revealing dynamic mechanisms of action. This evolution aims to address the significant uncertainty of the fate and effects of nanomaterials in real complex environmental scenarios.[85]

The starting point of the research is to establish quantifiable design criteria for the biosafety of nanomaterials. A systematic study was conducted on the toxicity of nanoparticles in embryonic zebrafish by integrating multiple ML algorithms and constructing a stacked ensemble model.[86] This work innovatively proposes an "EZ scoring system" that combines core surface properties such as zeta potential and aggregation size, providing key computational basis and screening criteria for the rational and safe design of nanomaterials.

Subsequently, the paradigm of toxicity prediction expanded from the individual biological level to the community level, which is closer to the real ecology. The study is the first to apply ML to predict the harmful concentration of engineering nanoparticles on aquatic communities. By comparing the performance of eight supervised learning models, a quantitative assessment of ecological risks at the community level has been achieved.[87] This model links the intrinsic properties of materials, such as sublimation enthalpy, with toxicity mechanisms, significantly improving the interpretability of prediction results and marking a crucial step toward the

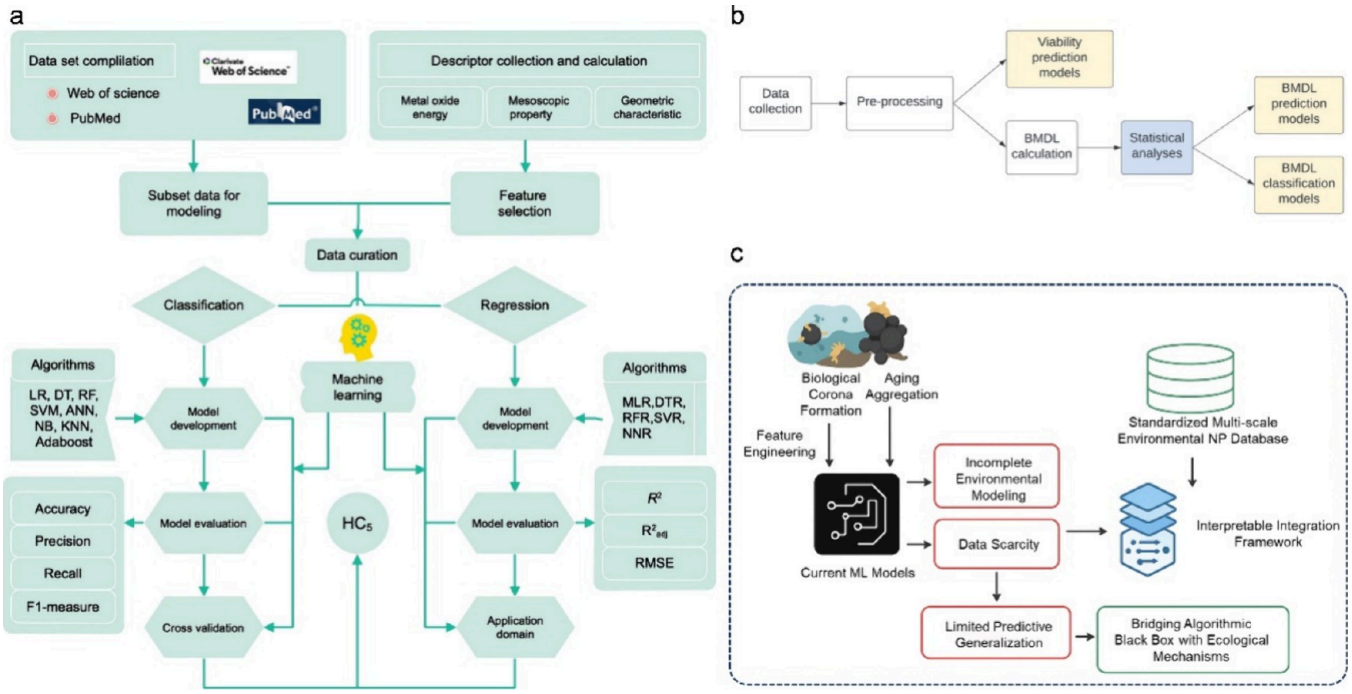


Figure 9. (a) Scheme for the prediction of HC5 based on ML methods. Reproduced with permission from ref 87. Copyright 2024, Springer Nature. (b) A schematic of the analysis strategy: data are collected and preprocessed; then, on one side models are fit on the data to predict the viability, on the other the data are used to calculate BMDL values, and the new data set is analyzed statistically, followed by fitting regression and classification models for the prediction of the BMDL or the BMDL range. Reproduced with permission from ref 89. Copyright 2022, Elsevier. (c) This chapter presents a schematic diagram of research work optimization and challenges.

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application of computational toxicology in ecosystem conservation (Figure 9a).

It is worth noting that different categories of nanomaterials have significant differences in their environmental toxicity mechanisms and key driving factors due to their unique physicochemical properties, which require prediction models to have material specificity. For example, in assessing the agricultural ecological risks of carbon dots, the random forest and gradient boosting models successfully quantified the nonlinear effects of factors such as particle size, dose, and emission spectra on plant toxicity responses.[88] On the other hand, a meta-analysis of graphene based materials reveals that their layer size, number of layers, and degree of surface functionalization are the dominant factors determining in vitro lung cell toxicity (Figure 9b).[89] These studies collectively indicate that in-depth feature engineering and targeted algorithm selection are crucial for building high-precision prediction models.

Although significant progress has been made in predictive performance in the current research, the common challenge is still insufficient modeling of the complexity of real environments (Figure 9c). Most existing models are difficult to effectively describe the dynamic transformation process of nanoparticles in environmental media, such as the formation, aging, and heterogeneous aggregation of biological crowns. To break this cyclic pattern, the new generation of models must incorporate environmental fate description indicators (such as conversion rate, heterogeneous aggregation tendency, and bioaccumulation coefficient) into key input features, thereby breaking free from the limitations of relying solely on intrinsic attributes and conventional toxicity indicators. Meanwhile, the scarcity of high-quality standardized data severely restricts the generalization ability of the model in different environmental scenarios. Looking ahead, there is an urgent need in this field to establish a standard database covering multiscale environmental processes and develop interpretable integrated computing frameworks (such as arbitration mechanisms based on SHAP values) to bridge the gap between the "algorithmic black box" and the understanding of ecological toxicology mechanisms, ultimately achieving accurate early warning and intelligent control of environmental risks in nanomaterials.

4.3. Bridge Across Biological Complexity

The field of cross species toxicity inference is undergoing a profound transformation with the help of ML methods, with the core goal of overcoming the mechanism barriers inherent in traditional extrapolation methods due to physiological and metabolic differences.[44,90−92] The current research paradigm is systematically constructing a more reliable multiscale prediction framework by establishing quantitative in vitro in vivo associations, utilizing model organisms as transformation bridges, and ultimately analyzing conserved molecular pathways.[14,93,94]

The first step in building a reliable cross species inference system is to establish a quantifiable correlation between in vitro biological responses and in vivo pathological end points. In response to the pulmonary toxicity of oxidized metal nanoparticles, a quantitative relationship between the levels of specific inflammatory cytokines in a cell model and mammalian pulmonary fibrosis injury was established for the first time by mining a large amount of experimental data. The prediction accuracy reached 80% in the test set.[95] This work validates the feasibility of converting in vitro molecular markers into complex pathological phenotypes in vivo, providing a key computational basis for subsequent preliminary screening and prediction of in vivo risks based on cellular levels.

Based on this quantitative correlation, selecting and standardizing a suitable live model organism as a bridge for transformation research are crucial for improving predicted flux and standardization. Zebrafish embryos, due to their in vitro development and optical transparency, have become an ideal model for connecting aquatic environments with mammalian systems. Related research has developed a high-throughput phenotype recognition system based on support vector machines, which can automatically analyze the color and texture features of embryos to quantify nanotoxicity, establishing a standardized and high-throughput phenotype data foundation for cross species inference.[96] Further research indicates that this prediction based on model organisms can be highly simplified. By constructing a lethal classification model, it was found that core physical and chemical parameters such as electronegativity are sufficient to effectively predict embryonic toxicity, suggesting that different species may share some basic toxicity response mechanisms.[97]

However, achieving truly robust cross species inference ultimately requires going beyond phenotypic associations and delving deeper into conservative molecular mechanisms. Breakthrough work has constructed hundreds of random forest models, successfully correlating the particle size, specific surface area, and other characteristics of titanium dioxide nanoparticles with a set of lung transcriptome biomarkers.[98] This study not only achieved cross scale prediction from nanophysical parameters to gene expression changes but also screened a conserved gene set in the toxicity process, providing a key mechanism anchor for identifying common molecular pathways across different species (Figure 10).

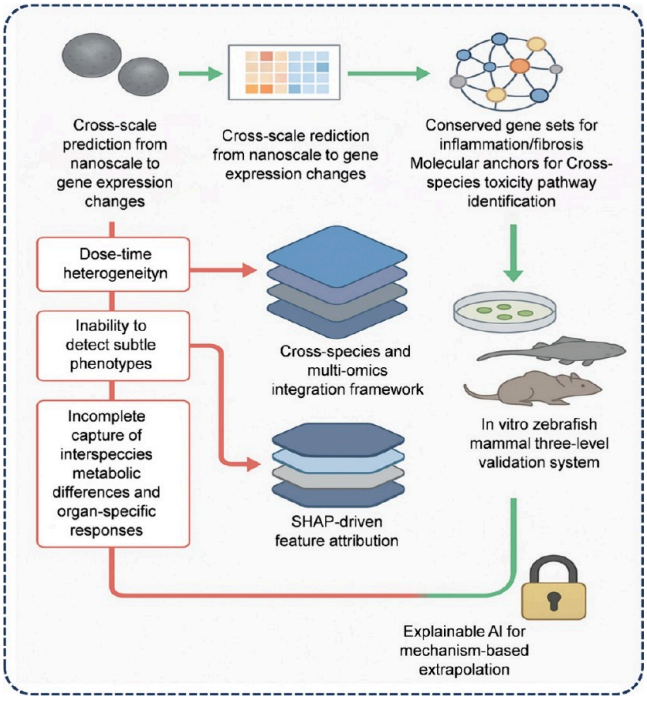


Figure 10. Current optimization and bottleneck diagram of ML in capturing cross species metabolic differences and organ specific responses.

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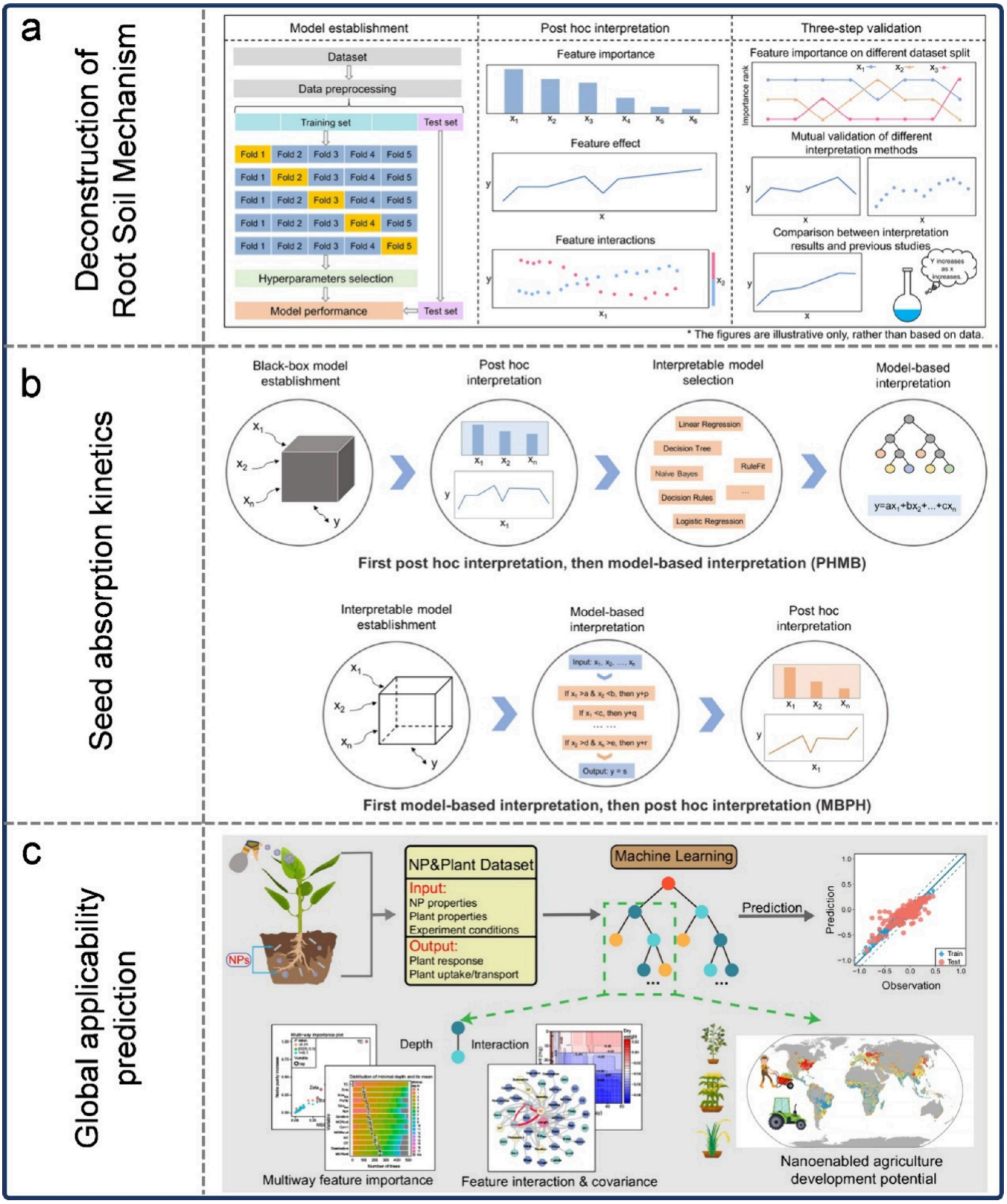


Figure 11. (a) Workflow of the model establishment and schematic illustration of the post hoc interpretation and three-step validation. Reproduced with permission from ref 104. Copyright 2022, The Royal Society of Chemistry. (b) Two kinds of integration approaches of post hoc and model-based interpretation. Reproduced with permission from ref 105. Copyright 2022, The Royal Society of Chemistry. (c) Workflow for studying the development potential of nanoenabled agriculture projected using ML with laboratory and field data. Reproduced with permission from ref 106. Copyright 2023, National Academy of Sciences.

Despite significant progress in the aforementioned research, the common limitation of current inference paradigms is still an insufficient analysis of the complexity of biological systems. For example, quantitative in vitro in vivo association models

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face standardization challenges due to inconsistent dose time responses; high-throughput phenotype systems have limitations in identifying subtle or sublethal phenotypes; and transcriptome based mechanism studies often lack integration of multi omics data to comprehensively reflect metabolic differences. The future development urgently requires the construction of a computational framework that integrates cross species and multiomics data and combines it with an in vitro zebrafish mammal stepwise experimental verification system. By applying interpretable AI technology to deeply analyze feature contributions, we expect it to ultimately solve the "black box" problem in cross species toxicity inference and achieve accurate and universal prediction of biological risks of nanomaterials.

5. AGRICULTURE AND FOOD TECHNOLOGY

The integration of AI and nanotechnology has not only significantly promoted the development of intelligent diagnosis and treatment technologies in the field of human health but also found extensive application areas in key challenges of agriculture and food safety. These fields serve as important practical experimental platforms for expanding and stress testing the core AI nanotechnology paradigm discussed earlier, particularly in the areas of nanosafety assessment and translational validation.[99] The research here directly faces the complexity of predicting the behavior and ecological impact of nanomaterials in open and complex environments, which is directly related to the assessment of nanomedicine safety in complex biological systems and provides a reference basis for the latter.

5.1. Intelligent Agricultural Nanomaterials

ML is profoundly reshaping the development paradigm of intelligent agricultural nanomaterials, with its core value being the ability to analyze the complex interactions of the "nanomaterials plant soil" system at multiple scales and achieve global predictions of material effects and agricultural applicability.[80,100−103] This intelligent process follows a clear path from analyzing micromechanics to verifying macro laws.

The starting point of the research is to establish a quantitative structure−activity relationship database for specific types of nanomaterials. By constructing a carbon dot plant interaction database covering over two thousand data points and applying random forest and gradient boosting models, early research systematically revealed the nonlinear regulatory laws of particle size, application dose, and luminescence characteristics on plant physiological response, providing the first data-driven design framework for customized synthesis of carbon dots for agricultural applications.[88]

After the basic structure−activity relationship was clarified, the research focus shifted to deciphering the behavior and effects of nanomaterials in more complex real agricultural environments, especially at the soil root interface. For this purpose, an interpretable ML framework combining LightGBM and RuleFit models has been developed. This framework employs a three-step validation strategy and found that in the soil root system, when the concentration of metal oxide nanoparticles exceeds a specific threshold, their root enrichment factor significantly decreases, challenging traditional dose-dependent linear cognition.[104] The same research framework has been extended to the process of seed germination, and its dual pathway explanatory strategy quantifies the synergistic effect between ion strength and nano zinc oxide, accurately locates the key concentration inflection points that affect the absorption dynamics of corn seeds, and elucidates the micro dynamics of the interaction between nanomaterials and biological interfaces from a mechanistic perspective (Figure 11a, b).[105]

The universality of the micromechanics revealed under controlled conditions needs to be verified and expanded on a larger spatial scale. By integrating over a thousand data sets worldwide and constructing multiple random forest models for analysis, the study systematically revealed the synergistic relationship between nighttime temperature and nanoparticle characteristics (such as size and surface charge) on crop growth.[106] This model confirms that nanoparticles with specific charges and small size characteristics are more effective in promoting growth in high-temperature areas and successfully predicts potential differences in the benefits of using nanoagricultural technology across different continents, thereby linking laboratory mechanisms with regional agricultural adaptability (Figure 11c).

The hierarchical analysis of agricultural complexity challenges as shown in Table 6, although significant progress has been made in these studies, the inherent biogeochemical complexity of agricultural systems poses fundamental challenges to ML models. Both micro mechanism models and macro prediction frameworks are generally limited by the lack of digitalization of key field environmental parameters, such as rhizosphere microbial activity or seed microstructure, and therefore often rely on simplified assumptions. At the same time, the insufficient integration of abiotic stress factors (such as drought) in existing models, as well as the uncertainty in predicting future extreme climate scenarios, highlight the urgent bottleneck of integrating heterogeneous data from multiple sources. It is necessary to develop a ML platform that integrates in situ rhizosphere sensing and multidimensional phenotype omics for future development in order to construct a fully interpretable intelligent prediction and optimization system that runs through the "nanomaterial properties environmental dynamics crop final phenotype".

Table 6. Hierarchical Analysis of Agricultural Complexity Challenges

Scale	Core missing parameters	Model limitations	Ref
Microscopic	Seed microcracks/ rhizosphere microbial activity	Simplification of seed absorption model	104
Field	Dynamics of drought/salt stress	The mechanism model does not cover environmental pressure	105
Regional—Global	Soil heterogeneity/ extreme climate events	Predict uncertainty	106

5.2. Food Safety Monitoring

The field of food safety monitoring is transitioning from passive sampling inspection to an active, real-time, and accurate risk warning paradigm through the integration of nano sensing technology and ML. The core of this transformation lies in the deep integration of the specific response of smart materials with the multimodal signal decoding ability of algorithms, in order to meet the high sensitivity and high specificity detection needs of various hazardous factors in complex food matrices.

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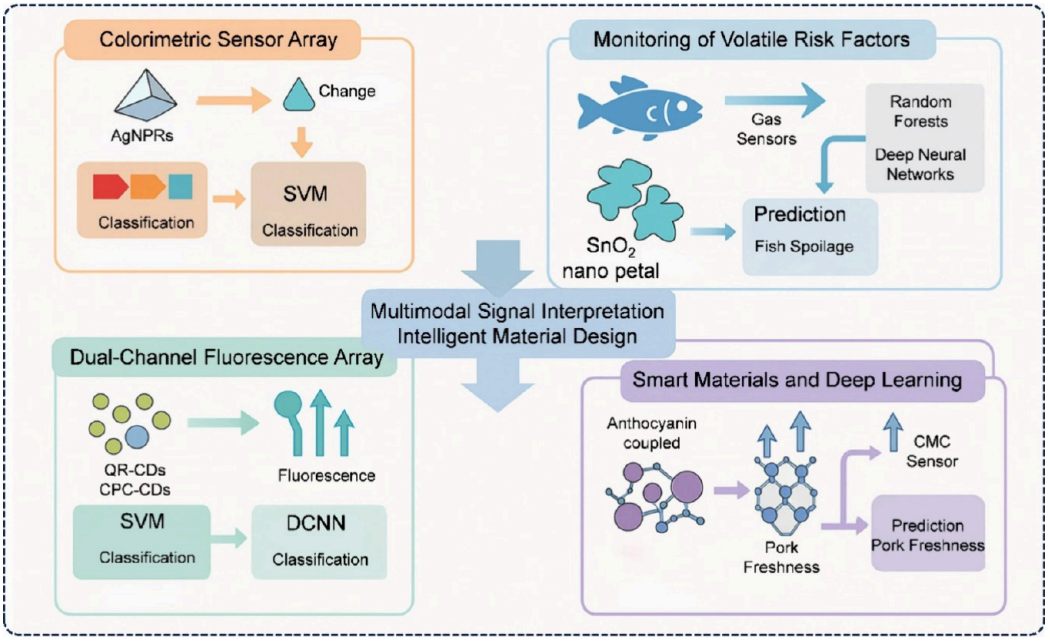


Figure 12. Overview of ML driven nanosensing technology.

Intelligent sensing has demonstrated its unique advantages in the digital evaluation of complex indicators such as food quality and flavor. Early research combined the etching kinetics of silver nanoprisms with indicator displacement effects to construct a dynamically responsive colorimetric sensor array.[107] Combined with support vector machine algorithm, this system not only achieves perfect identification of the origin of shrimp sauce but also quantitatively predicts its key flavor components, establishing the first framework based on dynamic chemical fingerprint for digital and standardized quality control of traditional fermented foods. Similar intelligent sensing logic is applied to monitor the process of food spoilage. Regarding the freshness of fish, a tin dioxide nano petal gas sensor was used to capture the decay marker formaldehyde, and combined with random forest and deep neural network, the identification of adulteration behavior and regression prediction of decay degree were achieved, effectively overcoming the subjectivity and lag of traditional sensory evaluation.[108]

For nonvolatile risk factors such as chemical pollutants, multichannel sensing strategies based on photophysical signals have shown strong potential. A fluorescence sensor array based on dual channel carbon quantum dots was developed, and its response fingerprints to different tetracycline antibiotics were decoded using a support vector machine. Accurate classification was successfully achieved in complex real-world samples such as river water and milk, confirming the value of multidimensional optical signals in improving detection specificity.[70]

Furthermore, cutting-edge research aims to integrate the structural and functional design of materials more closely with deep learning methods to enhance the environmental robustness and practicality of the system. By utilizing ionic liquids to regulate the color gamut of natural pigments and anchoring them onto metal organic framework nano frameworks, researchers have constructed mechanically stable intelligent sensing thin films.[109] The film, combined with deep convolutional neural networks, can accurately correlate the response of amine gases to the freshness level of pork. It is particularly noteworthy that the introduction of a nanosilica hydrophobic coating significantly improves the signal stability of the sensor in high humidity environments, reflecting the environmental adaptability design for practical applications.

However, to promote the practical application of these intelligent monitoring systems from the laboratory to cross scenarios, multiple challenges such as environmental robustness, long-term stability, and safety assessment still need to be addressed (Figure 12). For example, dynamic colorimetric sensing is susceptible to environmental light and temperature interference, gas sensors have signal drift problems, and the long-term stability of fluorescent sensing materials also needs to be improved. These all call for the development of adaptive algorithms with online self-calibration capabilities. Meanwhile, Table 7 compares the functional innovations of nanomaterials,

Table 7. Comparison of Functional Innovations of Nanomaterials

Material	Functional design	Ref
AgNPRs	Etching kinetics response	107
SnO_2 nano petals	Gas adsorption enhancement	108
Dual channel carbon quantum dots	Fluorescence fingerprint generation	70
ILs-UiO66@SiO_2	Color gamut tuning and environmental stability	109

and it can be seen that the potential migration risks of intelligent nanomaterials in food contact scenarios and their cross reactions to multicomponent interfering gases have not been systematically evaluated. Future development urgently needs to build an embedded edge computing architecture, which needs to integrate an in situ toxicity prediction module and combine with multimodal in situ detection technologies such as Raman fluorescence. The ultimate goal is to establish a closed-loop intelligent monitoring system that integrates environmental anti-interference, biosafety, and regulatory

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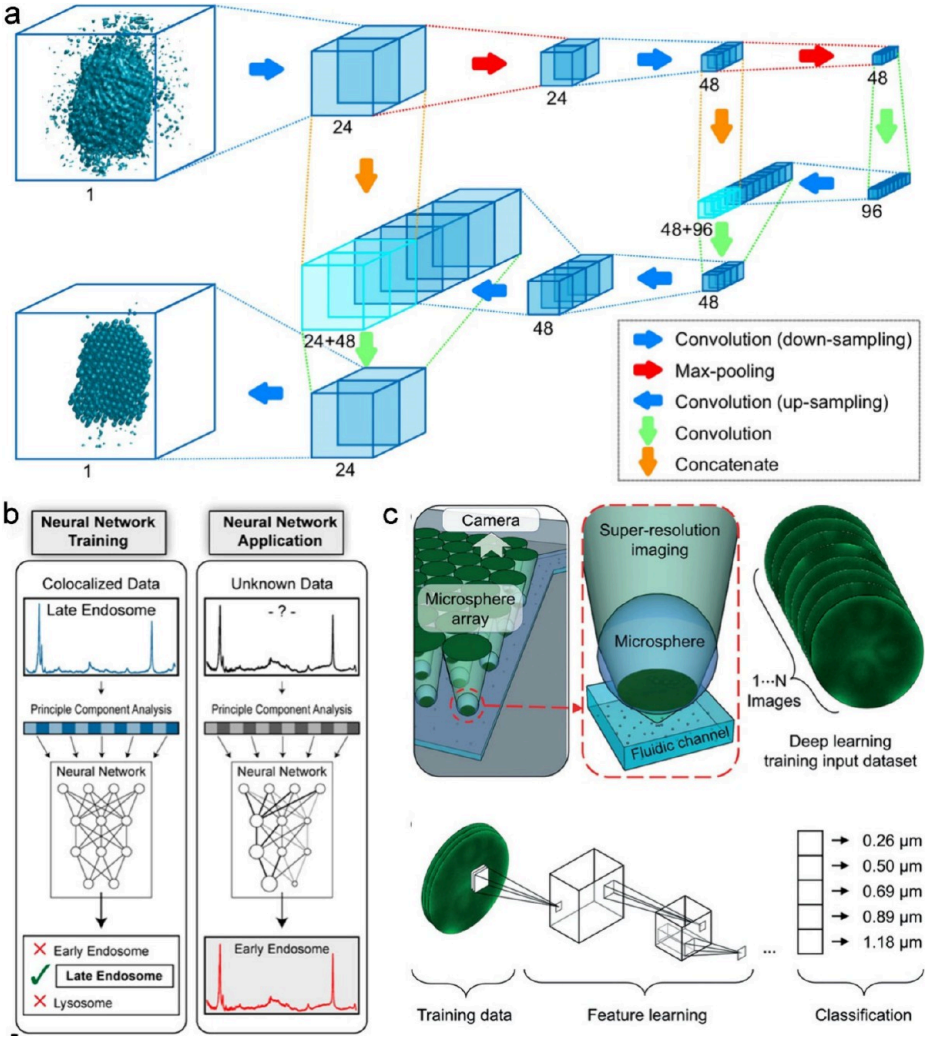


Figure 13. (a) The architecture of the deep learning augmentation. Reproduced with permission from ref 117. Copyright 2021, Nature. (b) Schematic depicting the general process of building and applying an artificial neural network. Reproduced with permission from ref 113. Copyright 2021, American Chemical Society. (c) (top) Schematic of the intelligent nanoscope setup, process, and data collection for deep learning training. (bottom) Schematic of the neural network process and nanoparticle classification. Reproduced with permission from ref 116. Copyright 2022, The Royal Society of Chemistry.

compliance requirements, laying the technical foundation for achieving full process intelligent control of food safety.

As mentioned earlier, the progress in the fields of intelligent agriculture and food safety monitoring fully demonstrates the powerful ability of the integration of AI and nanotechnology in solving specific industry challenges. However, the implementation of these complex applications fundamentally relies on a series of cross domain technical support. These core engines, including advanced data generation platforms, interpretable and reliable algorithms, and closed-loop development systems, are not only crucial for the field under discussion but also form the cornerstone of the entire ecosystem of AI and nanotechnology. The following sections will delve into the key supporting technologies that drive innovation from laboratory discoveries to practical applications.

6. TECHNICAL ENABLERS

6.1. Construction of Cross Scale Intelligent Data Generation Engine

The cross scale data generation technology based on ML is reshaping the observation paradigm of nanomedicine. Its core lies in integrating traditional fragmented atomic structures, molecular interactions, and cellular behavior information into a coherent analysis framework through intelligent imaging and spectroscopy methods.[18,110−114] This evolution aims to break through the physical limitations of a single technology and provide an unprecedented multidimensional, high-precision data foundation for understanding the complex interactions between nanomaterials and biological systems.

Accurate three-dimensional analysis of material structures at the atomic and nanoscale is the starting point for understanding their functions and biological effects. For this purpose, a neural network assisted atomic electron tomography technique has been developed, which can reconstruct the three-dimensional atomic arrangement on the surface of nanoparticles with picometer level accuracy.[115] Its deep learning driven data recovery strategy effectively compensates for the information loss caused by missing wedge-shaped data in traditional imaging, providing crucial atomic level inputs for subsequent quantum computing simulations of surface strain and electronic properties (Figure 13a).

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When the observation scale extends to the subcellular and molecular levels, dynamic, label-free microenvironment monitoring becomes a key requirement. A DNA functionalized single-walled carbon nanotube dual-mode optical probe was developed, which can simultaneously generate near-infrared fluorescence and Raman scattering signals.[113] By using ML to deconvolve the obtained multispectral fingerprints, real-time correlation analysis between parameters such as pH value and enzyme activity inside the endocytic vesicles and the aggregation state of nanotubes was achieved, providing an innovative tool for nondestructive monitoring of the fate of nanocarriers in living cells (Figure 13b). This recognition logic based on specific optical fingerprints can be further applied to the high-throughput classification of cell phenotypes. For example, by utilizing the stable near-infrared fluorescence features of specific DNA sequences and combining them with support vector machine algorithms, precise differentiation of M1/M2 polarization states of macrophages was successfully achieved, with a classification accuracy of up to 95%. A preliminary optical biomarker library for dynamic response of living cells has been established.[50]

In order to achieve high-throughput and high-throughput imaging analysis at the cellular and microbial scales, it is necessary to break through the diffraction limit and flux bottleneck of traditional optical microscopes. The development of intelligent nanomicroscopes provides a solution for this, which utilizes the photon nanojet effect of microsphere arrays to generate multiple super diffraction limit images in a single exposure.[116] After integration with deep convolutional neural networks, the system can perform high-throughput automatic classification of nanoparticles or bacteria at a speed of thousands of images per second, significantly improving the analysis efficiency and statistical reliability of microscopic samples (Figure 13c).

Although these technologies have made breakthroughs at their respective scales, building a fully integrated data generation engine that truly connects atoms, molecules, and cells still faces the common challenges of physical constraints and biological complexity. The future development urgently needs to develop an edge intelligence architecture that integrates "imaging spectroscopy omics". This architecture can combine physical information neural networks to constrain the physical authenticity of atomic reconstruction, enhance the generalization ability of spectral models in different biological scenarios using transfer learning, and embed microfluidic systems to achieve long-term dynamic monitoring of the cell in situ response. Through this systematic integration, it is expected to establish a standardized full chain data generation platform from quantum scale to living microenvironment (Table 8), providing a powerful engine for rational design and precise evaluation of nanomedicine.

6.2. Penetrating the "Double Black Box"

Explainable AI (XAI) refers to methods and techniques that make the outputs and decision-making processes of AI models understandable and interpretable to human experts, crucial for building trust and uncovering mechanistic insights in nanomedicine. XAI and multi task learning are becoming key technologies for analyzing the "material black box" and "biological black box" in the study of nanobiological interfaces.[48,95,118−120] By embedding prior knowledge of physical chemistry into algorithm frameworks and integrating biological features from different observation dimensions, these

Table 8. Summary of Core Breakthroughs in ML for Reshaping Molecular Cell Level Observation Paradigms in Nanomedicine through Intelligent Imaging and Spectral Fingerprint Analysis

Algorithm innovation	Breakthrough in data dimension	Ref
Restoration of missing wedge-shaped neural network	Three dimensional atomic displacement field (15 pm)	117
Multi spectral fingerprint deconvolution	Real time dynamic response of single-cell organelles	113
Microsphere image enhanced CNN classification	Thousand frames per second sub diffraction flux imaging	116

methods aim to establish transparent and interpretable prediction and understanding models ranging from nanostructures to complex biological responses.[7,121−125] It is crucial to distinguish between two complementary interpretability methods, each with its own advantages and disadvantages. Complementary use can better balance the predictive performance and interpretation needs. Although post hoc tools have flexibility, their explanations may be approximate and unstable. Models with inherent interpretability can provide a more solid causal foundation but may sacrifice some predictive performance when dealing with extremely complex and nonlinear problems.

The starting point for solving the "material black box" lies in combining first principles calculations with ML to provide a transparent computational paradigm for rational design of nanomaterials. Innovative research has integrated physical descriptors such as adsorption energy calculated by density functional theory into ML models, constructing a "transparent box" framework for interpretable prediction of two-dimensional nanoenzyme catalytic activity.[126] This method improves material screening efficiency by several orders of magnitude while maintaining quantum computing accuracy, providing clear design rules for designing high-performance nanoenzymes for specific applications, such as tumor microenvironments (Figure 14a).

At the cellular and biological response level, multi task learning frameworks are used to extract implicit information related to nano exposure from high-dimensional, heterogeneous biological data. For example, by using a random forest model to quantify unlabeled cell bright field images, dozens of morphological features were extracted and associated with nano toxicity reactions. Combined with dimensionality reduction and clustering analysis, the heterogeneity of cell populations and their relationship with functional phenotypes (such as vitality changes) were successfully revealed.[127] This multidimensional analysis from form to function achieves a more refined interpretation of nanobiological effects (Figure 14b). Furthermore, by combining high connotation imaging with deep learning, a more refined toxicity phenotype spectrum and mechanism of action can be established at the single-cell level. By applying multichannel cell imaging and convolutional neural networks, a detailed phenotype spectrum of nano material exposure was constructed at the single-cell scale, and matched with specific toxic mechanisms (such as interfering with photosynthetic electron transfer) through cluster analysis, achieving mechanism inference from complex phenotypes to clear biological pathways (Figure 14c).[128]

To ensure the reliability of the model interpretation, it has become particularly important to develop a cross validation framework that integrates multiple interpretation methods. The explanatory triangulation framework proposed for this

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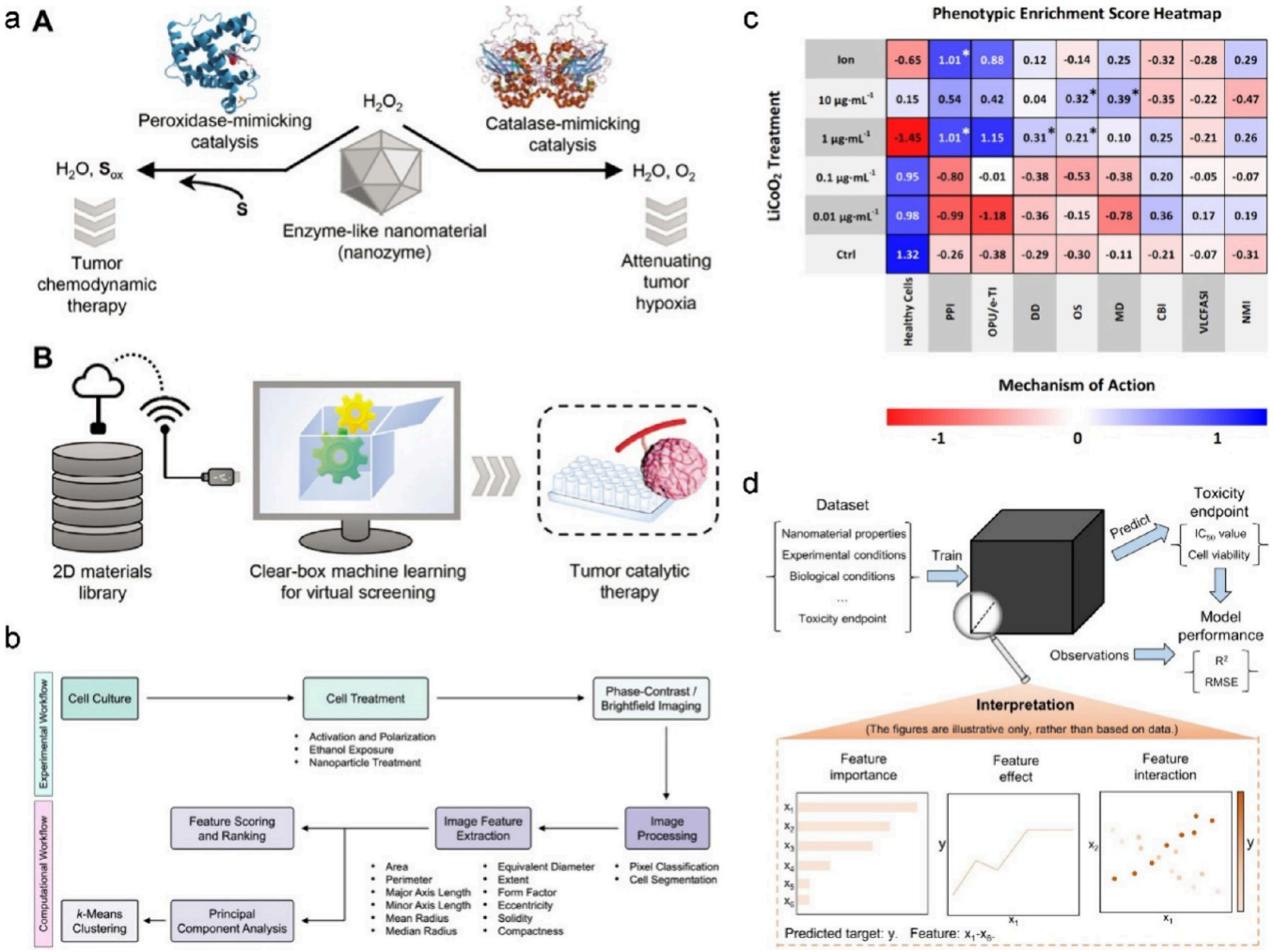


Figure 14. (a) Computational framework is proposed for the virtual screening. Reproduced with permission from ref 126. Copyright 2023, Wiley. (b) Overview of the experimental and computational workflows of the study. Reproduced with permission from ref 127. Copyright 2024, The Royal Society of Chemistry. (c) Phenotypic enrichment score heatmap. Reproduced with permission from ref 128. Copyright 2024, The Royal Society of Chemistry. d. The role of interpretation methods in ML models for nanotoxicity prediction. Reproduced with permission from ref 129. Copyright 2021, Elsevier.

purpose integrates multiple interpretable techniques such as SHAP, PDP, and ICE for systematic analysis of the mechanisms of quantum dot cytotoxicity.[129] This framework can decouple feature importance, effect direction, and interaction and transform the differences between different interpretation methods into tools for diagnosing data defects or model limitations, thereby enhancing the robustness of mechanism inference (Figure 14d).

The hybrid "glass box" model combines interpretable cores based on physical principles with transparent adaptive neural networks, enhancing the predictive ability while maintaining theoretical rigor. However, its development still faces the core challenge of balancing computational generalization and biological rationality. As shown in Table 9, existing research has demonstrated some key characteristics at the feature, model, and validation levels. Future breakthroughs require the use of new architectures such as neural symbol learning, integrating first principles, multi omics validation, and causal reasoning to construct differentiable causal models from microscopic properties to cellular pathways, thereby driving nanomedicine from empirical screening to mechanism oriented precision design.

Table 9. Summary of Synergy in Algorithm Innovation

Core Innovation	The Black Box Problem Solved	Ref
Adsorption energy descriptor driven cleaning model	Physical transparency of catalytic activity prediction	126
Morphological biomarkers without marker association	Quantitative bottleneck of cell heterogeneity	127
Phenotypic spectrum MoA matching framework	Nanotoxicity molecular pathway inversion	128
SHAP-PDP-ICE triangulation mutual verification	SHAP-PDP-ICE triangulation mutual verification	129

6.3. Calculation Experiment Closed-Loop System

The core value of the computational and experimental closed-loop system in the development of intelligent nanomaterials lies in bridging theoretical predictions and empirical feedback through ML, thereby transforming the traditional linear R&D model into an efficient iterative cycle of "design validation optimization".[61,130−133] The evolution of this paradigm is clearly reflected in the development path from optimizing specific manufacturing processes to rational design of molecular structures and then to building a full stack integrated platform.

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The early successful application of closed-loop systems is reflected in the multiobjective optimization of the complex nonlinear process of nanomaterial synthesis. By using a circular microreactor as a high-throughput experimental platform and the TSEMO algorithm as an intelligent optimization engine, the synergistic maximization of zinc oxide nanoparticle yield and antibacterial activity was successfully achieved.[1] This closed-loop strategy guided the completion of continuous synthesis on a kilogram scale in a very small number of experiments and simultaneously revealed the structure−activity relationship between nano morphology and performance, demonstrating its powerful effectiveness in solving practical production scaling problems.

When the research and development goals shift from process parameters to more complex molecular structure designs, closed-loop systems must incorporate deeper domain knowledge to address the challenge of data scarcity. In the optimization of near-infrared fluorescence performance of silver nanoclusters targeting DNA templates, the problem of data sparsity and class imbalance was effectively overcome by embedding crystal structure information as prior knowledge into ML feature engineering, achieving targeted functional design.[134] This work highlights the crucial role of chemical knowledge in enhancing model interpretability and predictive reliability, driving the evolution of closed-loop systems from parameter optimization to knowledge guided rational design (Figure 15a).

In order to cope with the exploration of large-scale combinatorial spaces, such as peptide sequences, the computational efficiency of closed-loop systems needs to be further upgraded through a layered strategy. The two-step active learning framework developed first uses a low precision model for initial screening of massive sequences, and then guides a high-precision model for fine iterative optimization of potential areas.[135] This hierarchical sampling strategy significantly reduces the computational burden and provides a scalable path for the high-throughput virtual screening of functional biological nanomaterials.

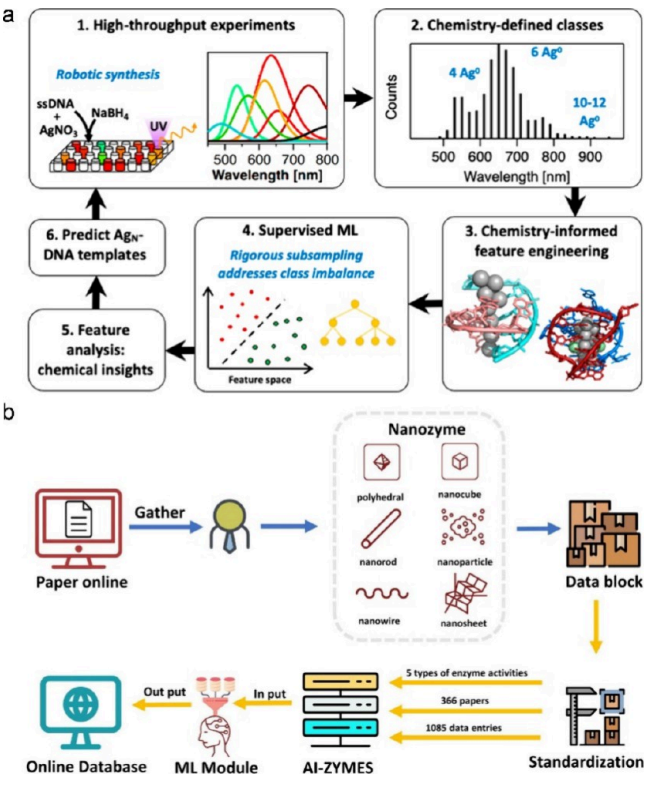


Figure 15. (a) Schematic of the workflow for ML-enabled AgN-DNA discovery (PDB accession codes 6NIZ40 and 6JR414). Reproduced with permission from ref 134. Copyright 2022, American Chemical Society. (b) Schematic overview of the Nanozyme database operation. Reproduced with permission from ref 136. Copyright 2025, Nature.

Ultimately, the mature form of closed-loop systems tends toward standardized and platform based integrated solutions. The constructed full stack platform AI-ZYMES integrates a nanoenzyme standard database, a dual-mode AI predictor, and a synthetic pathway generation module driven by a large language model.[136] This platform can quickly evaluate and design the catalytic performance of nanoenzymes, representing the development of closed-loop systems toward automation and cross domain collaboration (Figure 15b).

Looking at these studies, the evolution of the computation experiment closed-loop system shows a clear trajectory from single process optimization to multilevel rational design, high-throughput virtual screening, and platform integration development. Its core breakthrough lies in driving feature engineering through domain knowledge, actively learning to optimize experimental sampling, multiobjective Pareto front analysis, and standardized data architecture, greatly compressing the research and development cycle. However, moving toward broader applications still requires overcoming several challenges, such as degree of freedom limitations in complex process amplification, interpretability requirements for models under extreme data sparsity, and verification difficulties in multiscale mechanism coupling. Overcoming these challenges will be a key direction in promoting the next generation of intelligent closed-loop systems to better serve the efficient development of diagnostic and therapeutic nanomaterials.

7. TRANSLATIONAL MEDICINE CHALLENGE

7.1. Clinical Validation Bottlenecks

The core obstacle for intelligent nanomedicine platforms to move toward clinical applications lies in the insurmountable conversion gap between in vitro high-throughput prediction models and the highly dynamic and heterogeneous physiological environment in the human body.[137−142] Although recent research has attempted to bridge this gap by integrating multilevel data with innovative ML methods, the common challenges it reveals precisely highlight the complexity of the transformation path.

Initial efforts are focused on improving the accuracy of predicting specific pathological end points. For example, the multimodal ML framework constructed to predict metal oxide nanoparticle induced pulmonary fibrosis achieved high prediction accuracy, but its training set only contained a limited number of nanoparticles, making it difficult to fully simulate feature drift caused by dynamic protein crown formation and other processes in clinical environments.[103] In addition, the correlation between key in vitro biomarkers relied upon by the model and in vivo pathological phenotypes still needs to be validated through large-scale clinical samples (Figure 16a). This reflects the inherent limitations of models built on limited and static data sets in dealing with complex dynamic processes within the body.

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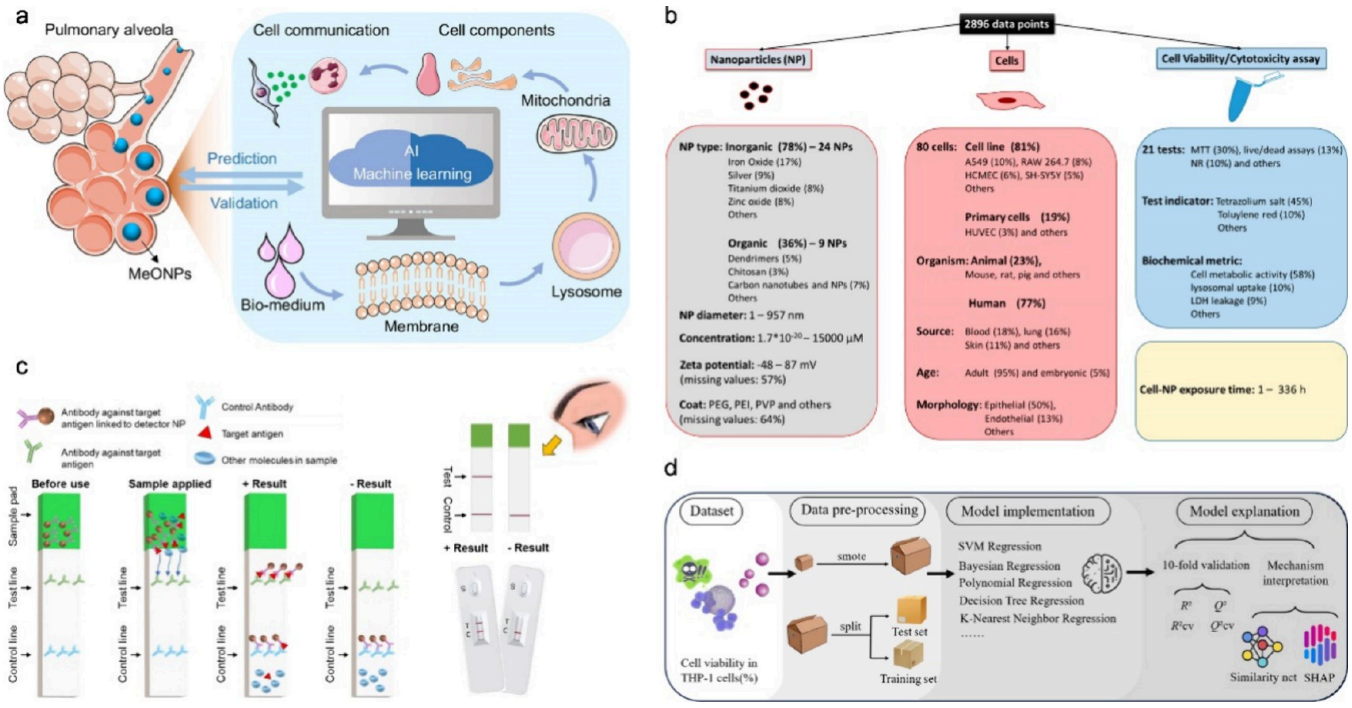


Figure 16. (a) Schematic workflow of multimodal feature fusion (MFF) modeling. Reproduced with permission from ref 103. Copyright 2025, Nature. (b) Overview of literature nanoparticle data attributes/features (2896 data points). Reproduced with permission from ref 143. Copyright 2019, American Chemical Society. (c) Schematic representation of a lateral flow assay (LFA) showing the sample pad containing the marked antibody against the target antigen and the test and control lines. Reproduced with permission from ref 133. Copyright 2024, Cold Spring Harbor Laboratory, USA. (d) Overall flowchart. ML models were trained and verified to obtain accurate prediction results. Reproduced with permission from ref 144. Copyright 2025, Elsevier.

In order to expand the coverage and universality of the model, another path is to integrate massive and diverse literature data. By integrating thousands of standardized cytotoxicity data points through meta-analysis, a decision tree model was developed to correlate the basic properties of materials (such as composition, size, charge) with toxicity effects.[143] Although this data-driven paradigm provides a valuable global perspective for the safe design of nanomaterials, the heterogeneity of source data (such as differences in cell types and exposure regimens) and the widespread lack of key features (such as surface modifications) severely limit the direct translational ability of the model in specific clinical toxicity assessments (Figure 16b).

At the same time, the development of low-cost materials for specific clinical testing applications also relies on the predictive ability of the models. The application of ML to evaluate alternative nanomaterials (such as titanium nitride and copper@gold core−shell structures) for side flow chromatography test strips has shown potential in improving detection sensitivity.[133] However, such models based on optimizing material properties often have significantly reduced prediction accuracy and generalization ability when faced with complex matrix interference in real clinical samples (Figure 16c). This points to a serious disconnect between the model validation environment and real-world application scenarios.

Further research attempts to enhance the robustness of the model by constructing a more balanced and multidimensional in vitro data set. By carefully constructing a data set containing multiple metal oxide nanoparticles and addressing the issue of class imbalance, research has improved the reliability of predicting macrophage toxicity.[144] However, even with more rigorous in vitro modeling methods, the fundamental limitation still lies in the inability to simulate the network effects of multi cell interactions in the human immune microenvironment, making it difficult to capture the real immune response dynamics in vivo (Figure 16d).

The above studies collectively reveal three core contradictions that hinder clinical translation, specifically reflected in insufficient data standardization, a lack of physiological relevance in models, and limited clinical adaptability of validation systems. To address these challenges, future research urgently needs to lead to breakthroughs in the following areas. First, it is necessary to develop characterization techniques that can capture the dynamic biological interface evolution of nanomaterials in situ. At the same time, a cross scale modeling method that integrates molecular, cellular, and tissue level mechanisms should be established. In addition, it is crucial to build a preclinical validation platform based on organoids or organ chips linked with AI prediction. Only through such a series of systematic efforts can intelligent nanoplatforms be transformed from powerful laboratory prediction tools to truly reliable clinical decision support systems.

7.2. Standardization and Repeatability

The standardization and reproducibility crisis faced by nanotoxicology research is rooted in the widespread data inconsistency, uncontrolled experimental variables, and lack of key structural annotations.[145−149] Recent research is addressing this challenge through systematic data governance and collaborative innovation, with a clear evolution path from identifying the root causes of variation to quantifying data quality and finally to achieving structural standardization.

The starting point of standardization efforts is to systematically identify and understand the main sources that lead to

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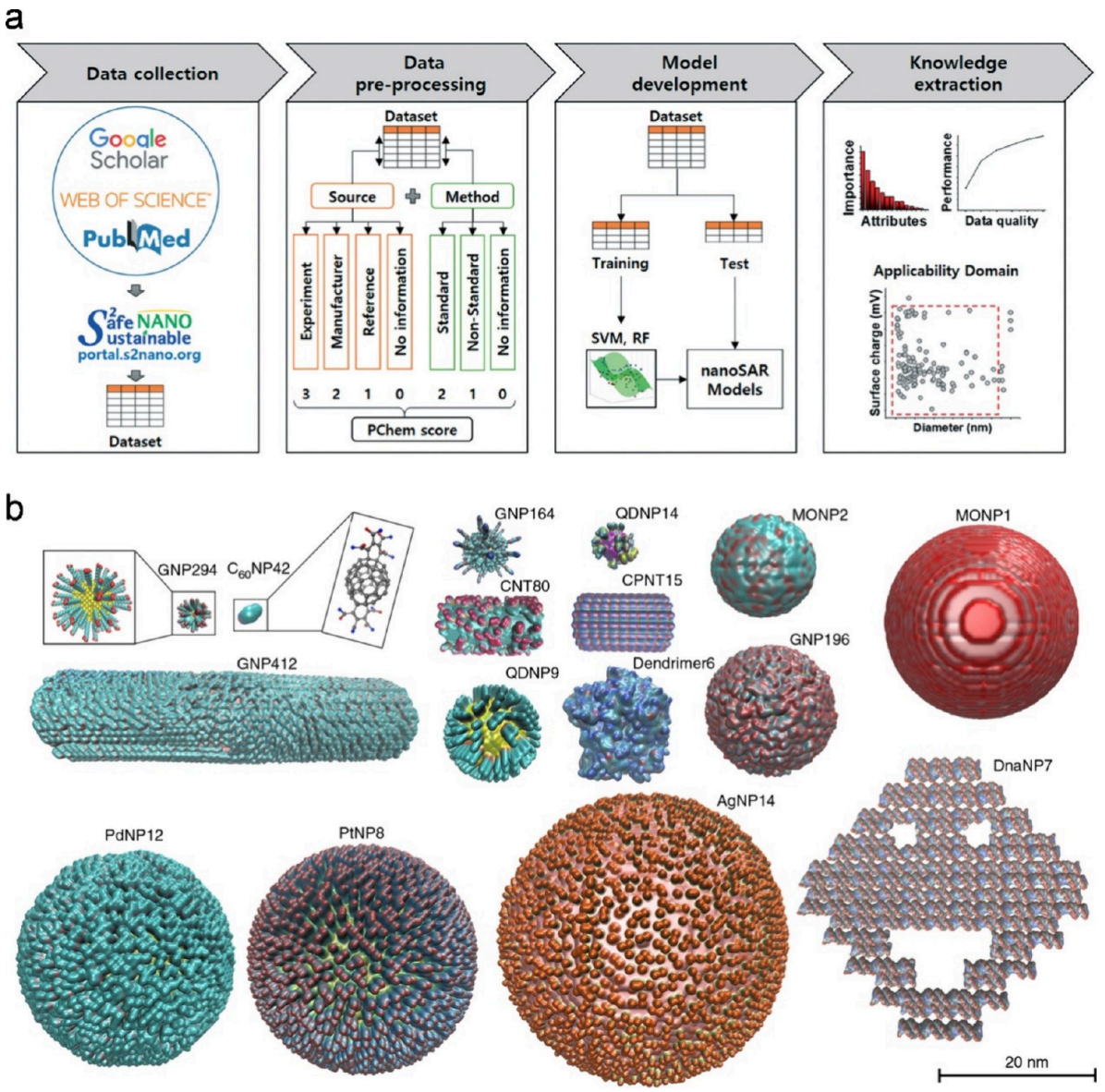


Figure 17. (a) Workflow for prediction model development. Reproduced with permission from ref 150. Copyright 2018, Royal Society of Chemistry. (b) Visualization of 16 representative nanomaterials in the database. Reproduced with permission from ref 91. Copyright 2020, Nature.

heterogeneity in the research results. Through meta-analysis of toxicity data of graphene materials, the study systematically identified that the differences between experimental conditions (such as exposure time, cell type) and inherent properties of the material (such as size, number of layers) are key factors leading to conclusions that cannot be repeated.[89] Although the semiquantitative structure−activity relationship model established from this has certain predictive ability, its stability is limited by the problem of uneven distribution of variables in the original data, which highlights the fundamental constraint of non standardized data on the generalization ability of ML models.

On the basis of identifying the root cause of the problem, establishing a quantifiable data quality assessment system has become a key step in promoting standardization. A PChem scoring system has been developed for quantitative quality assessment of nanotoxicology data in literature.[150] By rigorously screening historical data and constructing a graded quality subset, this study provides the first empirical evidence that using high-quality data can significantly improve the predictive performance of ML models, thereby establishing a quantitative correlation between data integrity and model reliability, providing a core tool for data governance (Figure 17a).

The higher-level stage of standardization lies in promoting the standardization and standardization of the characterization and annotation of nano material structures. The study solved the long-standing problem of atomic level structure annotation by unifying the structural information of hundreds of nanomaterials into a standard format for protein databases.[91] This measure not only supports the generation of automated computing descriptors but, more importantly, creates a benchmark platform for cross laboratory comparison, laying the foundation for achieving consistency and repeatability calculations at the structural level (Figure 7b).

The work presented above collectively outlines a standardized evolution path from "problem diagnosis" to "quality measurement" and then to "structural specification". In order

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to translate this conceptual progress into a concrete and actionable plan to improve the readiness of AI nanoplatforms for transformation, a community led, three-stage roadmap has been proposed. The initial "community adoption and foundation phase" focused on widely promoting and expanding existing minimum information standards, such as the OECD testing guidelines and ISO/TC 229 standards, to explicitly require data formats that are ML adapted. At the same time, establishing alliances to promote standardized data reporting is crucial, as it helps build a shared foundational layer. The next stage is the "technology integration phase", during which automated, ML assisted representation pipelines are developed and deployed, and large-scale, consistent data sets are generated, which is crucial for powerful model training. The final benchmarking and certification phase includes creating selected open-source benchmark data sets and international challenge competitions to objectively evaluate algorithm performance and material reproducibility. This phased approach provides an organized path from identifying problems to implementing scalable technical solutions and establishing performance benchmarks, thereby systematically building the infrastructure required for reliable translation. However, to meet the high standards of clinical translation, the existing system still faces three core challenges. The existing standards mainly focus on static physical and chemical properties, with insufficient coverage of dynamic biological processes in vivo, such as protein crown evolution; The data quality rating relies on historical literature and lacks a closed-loop feedback mechanism for mutual verification with controlled experimental models; Even if consistency in computational structure is achieved, there still exists a scale gap between in vitro standards and complex biological molecular interaction networks in vivo. The future breakthrough direction lies in developing new characterization standards that can capture dynamic interface processes, establishing a data validation framework that collaborates with physiological simulation platforms such as microfluidic organ chips and ultimately constructing a clinical grade nanomaterial identification system that integrates dynamic characteristics, quality scoring, standard structure, and multi-dimensional biological labels. This will ultimately achieve a fundamental leap from computational repeatability to clinical repeatability.

7.3. Regulatory Science and Innovation

The core challenge faced by regulatory science in the era of the integration of AI and nanotechnology is how to transform toxicity prediction models established at the laboratory scale into reliable and universal risk assessment frameworks that can cope with complex real-world environments and multi-component mixed exposure scenarios.[151−155] Recent research has attempted to drive regulatory paradigm innovation by constructing mechanism driven ML models and standardized data architectures, but its progress also reveals profound bottlenecks in ecological complexity and material universality.

Early innovation was reflected in expanding the traditional quantitative structure−activity relationship framework for a single substance to more practical risk assessment of nanomixtures. Related studies have integrated key theoretical descriptors such as the enthalpy of formation of gaseous cations, and constructed a dual model system that integrates support vector machines and neural networks, successfully achieving high-precision prediction of the toxicity of binary mixtures of various metal oxide nanoparticles.[156] The cross laboratory data compatibility protocol introduced in this work provides an important foundation for establishing regulatory standards for mixtures. However, the model also exposes blind spots in the current regulatory system. Its predictive ability is difficult to extrapolate to different categories such as carbon based nanomaterials, as the differences in toxicity mechanisms among different materials undermine the effectiveness of the unified descriptor system; Meanwhile, the model failed to consider the key feature drift caused by dynamic aggregation of nanoparticles in actual environments, which limits its practicality.

In order to extend the scale of risk assessment from laboratory test tubes to more closely resemble real ecosystems, subsequent research has developed ecological risk prediction models at the community level. This model constructs an ensemble learning framework by associating mechanism descriptors such as sublimation enthalpy and zeta potential with multi species toxicity end points, achieving excellent predictive performance.[87] The "mechanism anchoring algorithm integration" strategy adopted significantly improves the interpretability of the model and enhances its support value for regulatory decisions. However, this study also faces limitations. For example, the proportion of carbon based material samples in its training data is very low, and the model does not integrate the reshaping effect of water environment aging on the surface properties of nanoparticles, resulting in uncertainty in risk prediction for real ecological scenarios.

Overall, these two studies jointly outline the key path and ongoing challenges for regulatory scientific innovation. In order to follow this path and build a structured bridge from laboratory innovation to compliant products, it is necessary to align research and development work with constantly changing international regulatory concepts, such as the US Food and Drug Administration's (FDA) "Software as Medical Devices Based on AI /ML (SaMD) Action Plan" and the European Medicines Agency's (EMA) reflective document on nanomedicine. The key step is to adopt a customized Transition Readiness Level (TRL) framework for AI nanodiagnosis and therapy. This framework will clearly outline clear evidence requirements, starting from computational validation and computer simulation studies (TRL 1−3), progressing to experimental validation in biologically relevant environments (such as chip organ systems) (TRL 4−5), and ultimately conducting prototype testing and prospective pilot studies in simulated clinical environments to validate its robustness to individual differences (TRL 6−7). This structured and phased approach provides a practical roadmap from laboratory to clinical, encouraging early and sustained collaboration with regulatory agencies. In terms of methodology, there is an urgent need to establish a unified descriptor system that can span different material categories in order to move from single substance evaluation to multicomponent interaction research. In the verification system, it is necessary to combine cross laboratory standard protocols with verification platforms that can simulate in situ environmental processes (such as water aging) to bridge the gap between laboratory conditions and real ecological scenarios. In terms of regulatory applicability, it is urgent to establish a data extension mechanism driven by active learning to address the prediction blind spots of current models for emerging materials (such as carbon quantum dots for diagnosis and treatment) and iteratively optimize the prediction boundaries of the model by dynamically incorporat-

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ing high-risk material data. The future intelligent regulatory framework must integrate multiscale modeling covering molecular descriptors to ecological end points, modules that can dynamically modify environmental behavior (such as aggregation and aging), and adaptive architectures for different material categories, in order to ultimately achieve a paradigm shift from limited laboratory predictions to robust control of complex real-world risks.

8. SUMMARY AND OUTLOOK

Driven by the strong force of ML, the platform that integrates AI and nanotechnology is changing the overall way of disease diagnosis, precision treatment, and system safety assessment with unprecedented depth and breadth. This review systematically elucidates that ML enabled nanosensors have significantly improved the sensitivity and specificity of early diagnosis of cancer, infectious diseases, and other conditions; And based on algorithms such as reinforcement learning and graph neural networks, intelligent design closed loops have opened up new avenues for rational development and targeted delivery of nanomedicine; And interpretable AI models are gradually revealing the "black box" of nano biological interface interactions, providing a key tool for prospective assessment of their biological and environmental risks. However, there are still multiple conversion gaps between excellent laboratory "concept validation" and robust clinical "routine applications". The breakthrough progress in the future will rely on building intelligent, closed-loop, and convertible integrated system engineering.

8.1. Technological Frontiers and Enabling Paths for the Next Five to Ten Years

Specifically, the following emerging technologies and integration paths will dominate development in the next five to ten years:

(1) Computational paradigm: Neuromorphic computing can provide ultralow power, real-time response edge AI computing power for real-time detection and other scenarios, promoting the implementation of truly wearable intelligent diagnosis; Quantum ML is expected to unleash enormous potential in simulating complex nanomaterial systems and accelerating the discovery of functionally oriented new materials.

(2) Data and Model: Physical Information Neural Networks will integrate first principles and experimental observations to achieve multiscale accurate modeling from atoms to macroscopic levels; Federated learning can integrate clinical and experimental data from multiple centers and institutions while protecting data privacy, and solve the dilemma of data scarcity and isolation; Digital twin technology can construct patient specific nanomedicine efficacy prediction models, helping to achieve personalized medicine.

(3) Verification system: The highly biomimetic organ chip and human body chip system will become a key bridge connecting computational simulation, in vitro experiments, and clinical trials, producing high fidelity and standardized physiological and pathological response data, greatly improving the reliability of preclinical predictions.

(4) Closed loop system and clinical transformation: the "design verification optimization" intelligent closed loop deeply integrated with the above technologies, combined with the edge computing architecture embedded with the toxicity prediction module, will promote the realization of full process automation and real-time security early warning from "sample input" to "result output".

8.2. Deep Integration of Ethical, Social, and Regulatory Dimensions

With the leap of technological capabilities, the ethical, social, and governance challenges it raises must be examined synchronously and deeply:

(1) Data privacy and security: The massive patient multi omics and real-time physiological data generated and processed by the nanodiagnosis and treatment platform places extremely high demands on encrypted storage, secure transmission, and authorized access.

(2) Algorithm fairness and bias: If the training data fails to fully represent different races, genders, ages, and socioeconomic backgrounds, it may lead to systematic bias in diagnostic or prognostic algorithms, exacerbating health inequality.

(3) Interpretability and Responsibility Attribution: When AI systems participate in or even dominate critical diagnostic and treatment decisions, the non interpretability of the "black box" model can lead to serious legal and ethical responsibility delineation challenges.

(4) Fair accessibility and global health: It is necessary to ensure that these cutting-edge technologies not only serve resource rich regions, but also promote global health equity through technological innovation (such as developing low-cost, easy to deploy platforms) and policy design.

(5) Environmental sustainability: The full lifecycle environmental impact of smart nanomaterials, from synthesis, use to disposal, needs to be included in a standardized assessment framework to promote green design.

8.3. Cultivate an Innovative Ecosystem That Integrates Multiple Disciplines

To achieve the above vision, it is urgent to break down traditional disciplinary and organizational barriers and build an innovative ecosystem that is deeply integrated and co-evolving

(1) Interdisciplinary teams and integrated education: Establish interdisciplinary degree programs that integrate nanoscience, data science, and clinical medicine, and institutionalize the "transformation triangle" team composed of clinical doctors, material scientists, and algorithm engineers in major scientific research projects to promote demand-oriented innovation from the source.

(2) Open and collaborative data sharing platform: Under the framework of international cooperation, promote the establishment of standardized data infrastructure that follows the principles of discoverability, accessibility, interoperability, and reusability, support secure sharing and joint analysis of nanomaterials characterization, biological effects, and clinical outcome data, and crack data silos.

(3) Industry academia research medical management collaborative network: actively building a "regulatory sandbox" and public-private partnership, providing a secure space for collaborative innovation, resource exchange, and risk sharing among academia, industry,

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medical institutions, and regulatory authorities within a predetermined compliance framework, accelerating the transformation path from laboratory to product.

The integration of AI and nanotechnology is leading us toward a new paradigm of "design diagnosis treatment evaluation" that is intelligent, closed-loop, and people-oriented. By synergistically advancing in the three dimensions of technological frontier, ethical governance, and innovation ecology, the AI Nano fusion platform is expected to not only overcome the current transformation bottleneck but also profoundly reshape the future pattern of medical health, agricultural monitoring, and environmental governance, contributing core power to global sustainable health and development. Only through deep collaboration across disciplines, institutions, and departments can we truly unleash its transformative potential and achieve a leap from scientific innovation to social welfare.

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Notes

The authors declare no competing financial interest.

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